

Project Data Page:





Table of Contents

Cover Page	1
Vessel Design Summary:	2
Nozzle Summary:	5
Nozzle Schedule:	6
Bill of Material:	7
MDMT Summary:	8
Warnings and Errors:	10
Input Echo:	11
XY Coordinate Calculations:	18
Flg Calc [Int P]: CH. FLANGE	19
Flg Calc [Int P]: SH. FLANGE	25
Internal Pressure Calculations:	31
Element: CH. COVER	32
Element: CH. SHELL	34
Element: MAIN SHELL	36
Element: SH. COVER	38
Shop Pressure Test Results:	40
ASME TS Calc: Pressure Test	42
External Pressure Calculations:	50
Element and Detail Weights:	51
Nozzle Flange MAWP:	53
Wind Load Calculation:	54
Earthquake Load Calculation:	56
Center of Gravity Calculation:	58
Saddle Calcs: Bundle Pull New:	59
Saddle Calcs: Bundle Pull Cor:	60
Saddle Calcs: Operating Case:	61
Saddle Calcs: Test Case:	75
Nozzle Calcs.: TS IN NOZ. T1	88
Nozzle Calcs.: TS OUT NOZ. T2	98
Nozzle Calcs.: SS IN NOZ. S1	108
Nozzle Calcs.: SS OUT NOZ. S2	118
ASME TS Calc:	128



DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\A140NHT11QG001002-EQ161.pvdb

Date of Analysis : Sep 5, 2026 11:11am

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 660.000 mm. ID
 Vessel Design Length, Tangent to Tangent 5688.17 mm.
 Specified Datum Line Distance 50.00 mm.
 Shell Side Design Temperature 145 °C
 Channel Side Design Temperature 120 °C
 Shell Side Design Pressure 32.500 bars
 Channel Side Design Pressure 25.000 bars
 Shop Shell Side Test Pressure 44.046 bars
 Shop Channel Side Test Pressure 42.263 bars
 Wind Design Code ASCE/SEI 7-22
 Earthquake Design Code ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Flange	SA-266 2	...	...	K03506	
Nozzle	SA-266 2	...	...	K03506	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-179	...	...	K01200	
Tubesheet	SA-266 2	...	...	K03506	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
CH. COVER	25.065	0.00	54.000	3.0000	No	No
CH. SHELL	25.065	0.00	53.000	3.0000	N/A	No
CH. FLANGE	25.065	0.00	33.000	3.0000	N/A	No
SH. FLANGE	32.565	0.00	34.000	2.0000	N/A	No
MAIN SHELL	32.565	0.00	72.000	2.0000	N/A	No
SH. COVER	32.565	0.00	75.000	2.0000	No	No

Liquid Level: 660.00 mm. Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Ellipse	0.0	50.0	18.0	16.0	9.0	...	1.00	1.00
Cylinder	500.0	500.0	16.0	16.0	9.1	...	1.00	1.00
Body Flg	603.2	100.0	70.0	70.0	65.5	59.2	1.00	1.00
Body Flg	788.2	100.0	70.0	70.0	52.7	45.7	1.00	1.00
Cylinder	5588.2	4800.0	20.0	20.0	9.9	...	1.00	1.00
Ellipse	5638.2	50.0	23.0	20.0	9.8	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
TS IN NOZ. T1	WRC-297	0.868	Shell	Passed
TS OUT NOZ. T2	WRC-297	0.868	Shell	Passed
SS IN NOZ. S1	WRC-297	0.855	Shell	Passed
SS OUT NOZ. S2	WRC-297	0.855	Shell	Passed

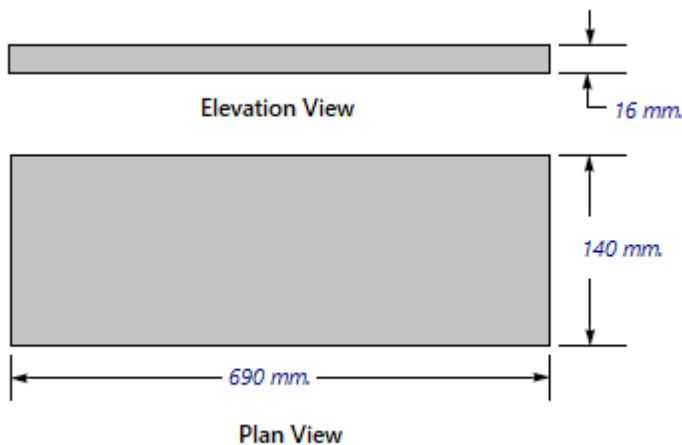
Foundation/Support Summary

Saddle Parameters:

Saddle Width	120.000 mm.
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	750.000 mm.
Wear Pad Width	160.000 mm.
Wear Pad Thickness	12.000 mm.
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	960.000 mm.

Baseplate Length	690.000 mm.
Baseplate Thickness	16.000 mm.
Baseplate Width	140.000 mm.
Number of Ribs (including outside ribs)	3
Rib Thickness	12.000 mm.
Web Thickness	12.000 mm.
Height of Center Web	306.000 mm.
Number of Bolts in Baseplate	2

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	6242.69 Kg
Maximum Transverse Saddle Shear Load	343.58 Kg
Maximum Longitudinal Saddle Shear Load	687.17 Kg

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	5384.47 Kg
Maximum Transverse Saddle Shear Load	102.01 Kg
Maximum Longitudinal Saddle Shear Load	32.56 Kg

Weights:

Fabricated - Bare w/o removable internals	5656.7 kg.
Shop Test - Fabricated + water (full)	7416.1 kg.
Shipping - Fab. + rem. intls.+ shipping app.	5656.7 kg.
Erected - Fab. + rem. intls.+ insul. (etc)	5656.7 kg.

PV Elite 28 Licensee: SL User
FileName : A140NHT11QG001002-EQ161
Vessel Design Summary:

Step: 29 11:11am

Page 4 of 135
Sep 5, 2026



HEXAGON

Empty	- Fab. + intls. + details + wghts.	5656.7 kg.
Operating	- Empty + operating liquid (no ca)	7416.1 kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
TS IN NOZ. T1	39	...	...	OK	9.12	OK	Passed
TS OUT NOZ. T2	39	...	...	OK	9.10	OK	Passed
SS IN NOZ. S1	44	...	...	OK	9.95	OK	Passed
SS OUT NOZ. S2	44	...	...	OK	9.93	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 44 Nozzle : SS IN NOZ. S1 [Shellside]
 Minimum MAWP Nozzles : 39 Nozzle : TS IN NOZ. T1 [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm.	Layout Angle deg	Dia. Limit mm.
20	TS IN NOZ. T1	300.000	-90.000	418.400
20	TS OUT NOZ. T2	300.000	90.000	418.400
50	SS IN NOZ. S1	1238.175	-90.000	820.800
50	SS OUT NOZ. S2	5138.175	90.000	820.800

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
TS IN NOZ. T1	8.000 in	Actual	WNF	9.000	12.700	...	...	241.25	600
TS OUT NOZ. T2	8.000 in	Actual	WNF	9.000	12.700	...	...	241.25	600
SS IN NOZ. S1	16.000 in	Actual	WNF	17.000	12.700	...	...	311.62	300
SS OUT NOZ. S2	16.000 in	Actual	WNF	17.000	12.700	...	...	261.62	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
TS IN NOZ.	SA-266 2	16.000	10.000	...	...	...
TS OUT NOZ.	SA-266 2	12.000	10.000	...	...	...
SS IN NOZ.	SA-266 2	16.000	10.000	...	...	...
SS OUT NOZ.	SA-266 2	16.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
TS IN NOZ. T1	250.000	-90.0	200.00	0.00	CH. SHELL
TS OUT NOZ. T2	250.000	90.0	200.00	0.00	CH. SHELL
SS IN NOZ. S1	1188.175	-90.0	200.00	0.00	MAIN SHELL
SS OUT NOZ. S2	5088.175	90.0	150.00	0.00	MAIN SHELL

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	ELLIPTICAL HEAD: 2.0 X 1, 18.0mm. THK X 660.0mm. ID X 50.0mm.	SA-516 70
1	CYLINDER: 16.0mm. THK X 660.0mm. ID X 500.0mm.	SA-516 70
2	WELD NECK FLANGE: 70.0mm. THK X 835.0mm. OD	SA-266 2
1	CYLINDER: 20.0mm. THK X 660.0mm. ID X 4800.0mm.	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 23.0mm. THK X 660.0mm. ID X 50.0mm.	SA-516 70
9	BAFFLES: 648.0mm DIA X 8.0mm THK	SA-516 70
6	TIE RODS: 12mm DIA X 4600mm	SA-193 B7M
6	SPACERS: 20.0mm DIA X 1.5mm THK X 4600.0mm	SA-179
1	TUBESHEET: 835.0mm X 85.0mm THK	SA-266 2
442	TUBES: 4600.0mm X 19.0mm DIA X 2.1mm THK	SA-179
2	SADDLE: 120.0mm X 120 DEG	SA-516 70
1	GASKET: 718.0mm. OD X 692.0mm. ID	...
32	BODY FLANGE BOLTS: 24.0mm. DIA	SA-193 B7
64	NUTS FOR BODY FLANGE BOLTS: 24.0mm. DIA	...
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
SH. FLANGE [11]		B	-7	-44	-29	0.468	20.000	1.00	No
MAIN SHELL [8]		B	-7	-46	-29	0.460	20.000	1.00	No
SH. COVER [10]		B	-7	-47	-29	0.450	20.000	1.00	No
SH. COVER [7]		B	-3	-48	-29	0.394	23.000	1.00	No
FIXED SADDLE [24]		B	-23	-48	-29	0.460	12.000	1.00	No
RHS Saddle [24]		B	-23	-48	-29	0.460	12.000	1.00	No
SS IN NOZ. S1 [1]		B	-1	-39	-29	0.459	12.700	1.00	No
Nozzle Flg [5]		A	-18	-38					
SS OUT NOZ. S2 [1]		B	-1	-40	-29	0.458	12.700	1.00	No
Nozzle Flg [5]		A	-18	-38					
Tubesheet: SS [13]		B	-6	-8	-29	0.957	21.250	1.00	No
Warmest MDMT:			-1	-8					
CH. FLANGE [11]		B	-14	-35	-29	0.618	16.000	1.00	No
CH. COVER [10]		B	-14	-37	-29	0.598	16.000	1.00	No
CH. COVER [7]		B	-11	-39	-29	0.531	18.000	1.00	No
CH. SHELL [8]		B	-14	-36	-29	0.612	16.000	1.00	No
TS IN NOZ. T1 [1]		B	-14	-48	-29	0.470	16.000	1.00	No
Nozzle Flg [5]		A	-18	-104					
TS OUT NOZ. T2 [1]		B	-14	-48	-29	0.469	16.000	1.00	No
Nozzle Flg [5]		A	-18	-104					
Tubesheet: CS [14]		B	-6	-8	-29	0.957	21.250	1.00	No
Bolting [21]			-48						
Warmest MDMT:			-6	-8					
Exchanger Side		Computed MDMT		Required MDMT		Pass/Fail			
Shell		-29.0		-28.9		Pass			
Channel/Tube		-29.0		-28.9		Pass			

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.



UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

Warning: CH. FLANGE and SH. FLANGE

There are mating flanges in this model that have different design pressures. This will generate different bolt loads. If a tubesheet is in between these flanges, PV Elite will compute the mating flange loads for you. If there is not a tubesheet physically separating these flanges, you must enter in the mating flange loads into the flange dialog.

Warning:

There is a flange in this model connected to an element whose temperature is different. The bolt allowable stress is based on it's parent design temperature. You may need to manually adjust the operating bolt allowable stress to a value based upon a representative maximum temperature the bolts will be subject to.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (Kgfcmm2mm):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	32.5	bars
Channel Side Design Pressure	25	bars
Shell Side Design Temperature	145.0	°C
Channel Side Design Temperature	120.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Sour	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	-28.9	°C
MDMT (CET), Tube Side	-28.9	°C
User defined MAWP, Shell Side	0	bars
User defined MAWP, Channel Side	0	bars
User defined MAPnc, Shell Side	0	bars
User defined MAPnc, Channel Side	0	bars
User defined Test Pres., Shell Side	0	bars
User defined Test Pres., Channel Side	0	bars

Projection of Nozzle from Vessel Top	0	mm.
Projection of Nozzle from Vessel Bottom	0	mm.
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS



Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 Km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	0
Base Elevation	300 mm.
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment	H or Hh 0 mm.
Distance Upwind of Crest	Lh 0 mm.
Distance from Crest to the Vessel	x 0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Seismic Design Code	ASCE/SEI 7-22
Importance Factor	1.500
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio	z/h 0.000
Amplification Factor	Ap 0.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Component resonance duct. fact.	[CAR] 0.000
Component strength factor	[Rpo] 0.000
Overstrength factor	[Omega0] 0.000
Response Acceleration Factor	[SMS] 0.326
Response Acceleration Factor	[SM1] 0.139
Building Importance Factor	[Ie] 1.000
Site Latitude	0.000 deg
Site Longitude	0.000 deg

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No



Use the MAWP to compute the MDMT Yes
 For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes
 For PWHT & P1 Materials the MDMT can be < -55F (-48C) No

 Using Metric Material Databases, ASME II D Yes
 Calculate B31.3 type stress for Nozzles with Loads Yes
 Reduce the MDMT due to lower membrane stress Yes
 Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/6	
Element From Node	10	
Element To Node	20	
Element Type	Elliptical	
Description	CH. COVER	
Distance "FROM" to "TO"	50	mm.
Inside Diameter	660	mm.
Element Thickness	16	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	18	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	25	bars
Design Temperature Internal Pressure	120	°C
Design External Pressure	0	bars
Design Temperature External Pressure	120	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	234	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm.
Yield Stress, Operating	235.8	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Liquid	
Detail ID	Liquid 10	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	660	mm.
Liquid Density	0.0009984	kg./cm ³

Element#	2/6	
Element From Node	20	
Element To Node	30	
Element Type	Cylinder	
Description	CH. SHELL	
Distance "FROM" to "TO"	500	mm.
Inside Diameter	660	mm.
Element Thickness	16	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	16	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	25	bars
Design Temperature Internal Pressure	120	°C
Design External Pressure	0	bars
Design Temperature External Pressure	120	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	



Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	20	
Detail Type	Liquid	
Detail ID	Liquid: 20	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	660	mm.
Liquid Density	0.0009984	kg./cm ³
Element From Node	20	
Detail Type	Nozzle	
Detail ID	TS IN NOZ. T1	
Dist. from "FROM" Node / Offset dist	250	mm.
Nozzle Diameter	8	in.
Nozzle Schedule	80	
Nozzle Class	600	
Layout Angle	-90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	74.763	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	TS OUT NOZ. T2	
Dist. from "FROM" Node / Offset dist	250	mm.
Nozzle Diameter	8	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	74.763	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	

Element#	3/6	
Element From Node	30	
Element To Node	40	
Element Type	Flange	
Description	CH. FLANGE	
Distance "FROM" to "TO"	100	mm.
Flange Inside Diameter	660	mm.
Element Thickness	70	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	70	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	25	bars
Design Temperature Internal Pressure	120	°C
Design External Pressure	0	bars
Design Temperature External Pressure	120	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	179.4	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm.
Yield Stress, Operating	223.8	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	30
Detail Type	Liquid



Detail ID	Liquid 30
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	660 mm.
Liquid Density	0.0009984 kg./cm ³

Element#	4/6
Element From Node	40
Element To Node	50
Element Type	Flange
Description	SH. FLANGE
Distance "FROM" to "TO"	100 mm.
Flange Inside Diameter	660 mm.
Element Thickness	70 mm.
Internal Corrosion Allowance	2 mm.
Nominal Thickness	70 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	32.5 bars
Design Temperature Internal Pressure	145 °C
Design External Pressure	0 bars
Design Temperature External Pressure	120 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	40
Detail Type	Liquid
Detail ID	Liquid 40
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	660 mm.
Liquid Density	0.0009984 kg./cm ³

Element#	5/6
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	MAIN SHELL
Distance "FROM" to "TO"	4800 mm.
Inside Diameter	660 mm.
Element Thickness	20 mm.
Internal Corrosion Allowance	2 mm.
Nominal Thickness	20 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	32.5 bars
Design Temperature Internal Pressure	145 °C
Design External Pressure	0 bars
Design Temperature External Pressure	120 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	179.4 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm.
Yield Stress, Operating	262 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Saddle
Detail ID	FIXED SADDLE



Dist. from "FROM" Node / Offset dist	960	mm.
Width of Saddle	120	mm.
Height of Saddle at Bottom	750	mm.
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm.
Width of Wear Plate	160	mm.
Thickness of Wear Plate	12	mm.
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.60000002	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm.
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm.
Element From Node	50	
Detail Type	Saddle	
Detail ID	RHS Saddle	
Dist. from "FROM" Node / Offset dist	3840	mm.
Width of Saddle	120	mm.
Height of Saddle at Bottom	750	mm.
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm.
Width of Wear Plate	160	mm.
Thickness of Wear Plate	12	mm.
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.30000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm.
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm.
Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid 50	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	660	mm.
Liquid Density	0.0009984	kg./cm ³
Element From Node	50	
Detail Type	Nozzle	
Detail ID	SS IN NOZ. S1	
Dist. from "FROM" Node / Offset dist	400	mm.
Nozzle Diameter	16	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	-90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	153.98	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	
Element From Node	50	
Detail Type	Nozzle	
Detail ID	SS OUT NOZ. S2	
Dist. from "FROM" Node / Offset dist	4300	mm.
Nozzle Diameter	16	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	169.42	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	



Element To Node	70	
Element Type	Elliptical	
Description	SH. COVER	
Distance "FROM" to "TO"	50	mm.
Inside Diameter	660	mm.
Element Thickness	20	mm.
Internal Corrosion Allowance	2	mm.
Nominal Thickness	23	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	32.5	bars
Design Temperature Internal Pressure	145	°C
Design External Pressure	0	bars
Design Temperature External Pressure	120	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	60	
Detail Type	Liquid	
Detail ID	Liquid 60	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	660	mm.
Liquid Density	0.0009984	kg./cm ³



XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
CH. COVER		50	...	50	...
CH. SHELL		550	...	500	...
CH. FLANGE		653.175	...	100	...
SH. FLANGE		838.175	...	100	...
MAIN SHELL		5638.17	...	4800	...
SH. COVER		5688.17	...	50	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

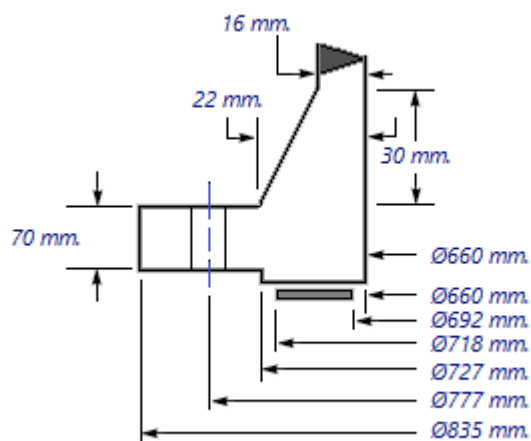
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Flange Input Data Values

Description: CH. FLANGE :

CH. FLANGE

Flange Type	Integral Weld Neck		
Design Pressure	P	25.06	bars
Design Temperature		120	°C
Internal Corrosion Allowance	ci	3.0000	mm.
External Corrosion Allowance	ce	0.0000	mm.
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	660.000	mm.
Flange Outside Diameter	A	835.000	mm.
Flange Thickness	t	70.0000	mm.
Thickness of Hub at Small End	go	16.0000	mm.
Thickness of Hub at Large End	gl	22.0000	mm.
Length of Hub	h	30.0000	mm.
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	777.000	mm.
Nominal Bolt Diameter	a	24.0000	mm.
Type of Threads		TEMA Metric	
Number of Bolts		32	
Flange Face Outside Diameter	Fod	727.000	mm.
Flange Face Inside Diameter	Fid	660.000	mm.
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	718.000	mm.
Gasket Inside Diameter	Gi	692.000	mm.
Gasket Factor	m	3.7500	
Gasket Design Seating Stress	y	62.06	N./mm ²
Gasket Thickness	tg	3.1750	mm.
Raised face depth	Fd	0.0000	mm.
Length of Partition Gasket	lp	646.0000	mm.
Width of Partition Gasket	tp	13.0000	mm.
Partition Gasket Factor	mPart	3.7500	
Partition Gasket Design Seating Stress	yPart	62.06	N./mm ²





ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (25.06 \cdot (660.0/2 + 3.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 25.06) + Ca$$

$$= 9.1153 \text{ mm.}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 13.0) / (333.0 + 0.6 \cdot 13.0)$$

$$= 52.637 \text{ bars}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 16.0) / (330.0 + 0.6 \cdot 16.0)$$

$$= 65.013 \text{ bars}$$

Corroded Flange ID,	Bcor = B+2*ci	666.000 mm.
Corroded Large Hub,	g1Cor = g1-ci	19.000 mm.
Corroded Small Hub,	g0Cor = go-ci	13.000 mm.
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	36.500 mm.
Gasket Contact Width,	N = (Go - Gi)/2	13.000 mm.
Basic Gasket Width,	bo = N/2	6.500 mm.
Effective Gasket Width,	b = Cb*sqrt(bo)	6.425 mm.
Gasket Reaction Diameter,	G = Go - 2 * b	705.151 mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 705.1509^2 \cdot 25.065$$

$$= 99818.945 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot lp \cdot bPart \cdot mPart \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 705.1509 \cdot 3.75 \cdot 25.06$$

$$+ 2 \cdot 646.0 \cdot 6.4246 \cdot 3.75 \cdot 25.0647$$

$$= 35239.262 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 666.0^2 \cdot 25.0647$$

$$= 89042.500 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 99819 - 89043$$

$$= 10776.444 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(99819 + 35239 + 0, 0)$$

$$= 135058.203 \text{ Kg}$$

$$= 168381.672 \text{ Kg, Mating Flange Load Governs}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 62.06 \cdot 6.4246 \cdot \pi \cdot 705.151 + 62.06 \cdot 6.4246 \cdot 646.0$$

$$= 116320.086 \text{ Kg}$$

$$= 126326.547 \text{ Kg, Mating Flange Load Governs}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((168382 + 0 + 4 \cdot 0 / 705.15) / 172, 126327 / 172)$$

$$= 96.007 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 24.0 + 6 \cdot 70.0 / (3.75 + 0.5)$$



= 146.824 mm.

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 777.0 \cdot \sin(\pi / 32)$$

$$= 76.159 \text{ mm.}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs / (2a + t)}, 1)$$

$$= \max(\sqrt{76.159 / (2 \cdot 24.0 + 70.0)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	96.007	100.079	
Radial Distance between Hub and Bolts:	33.500	36.500	
Radial Distance between Bolts and Edge:	29.000	29.000	
Circ. Spacing between the Bolts:	57.500	76.159	146.824

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 100.079 \cdot 172.0 / (62.06 \cdot \pi(718.0 + 692.0))$$

$$= 6.262 \text{ mm.}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 16 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 100.0794$$

$$= 175523.97 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 168382 - 99819$$

$$= 68562.73 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (777.0 - 705.1509) / 2$$

$$= 35.9245 \text{ mm.}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((777.0 - 666.0) / 2 + 35.9245)$$

$$= 45.7123 \text{ mm.}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (777.0 - 666.0 - 19.0) / 2$$

$$= 46.0000 \text{ mm.}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	89042.	46.0000	1.0000	4096.
Face Pressure, Mt	10776.	45.7123	1.0000	493.
Gasket Load, Mg	68563.	35.9245	1.0000	2463.
Gasket Seating, Matm	175524.	35.9245	1.0000	6306.

Total Moment for Operation, Mo 7052. Kg-m.
 Total Moment for Gasket seating, Mg 6306. Kg-m.

Effective Hub Length, $h_o = \sqrt{Bcor \cdot goCor}$ 93.048 mm.
 Hub Ratio, $h/h_o = HL / H_o$ 0.322
 Thickness Ratio, $g_1/g_o = (g_1Cor/goCor)$ 1.462

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.879
 Factor V per 4.16.5 0.371
 Factor f per 4.16.5 1.047

Stress factor from Table 4.16.4 [K]:

$= A / B$
 $= 835.0/666.0$
 $= 1.254$

Stress factor from Table 4.16.4 [Y]:

$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$
 $= (0.66845 + (5.7169(1.254^2 \cdot \log_{10}(1.254))) / (1.254^2 - 1)) / (1.254 - 1)$
 $= 8.716$

Stress factor from Table 4.16.4 [T]:

$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 $= (1.254^2(1 + 8.55246 \cdot \log_{10}(1.254)) - 1) / ((1.0472 + 1.9448 \cdot 1.254^2) \cdot (1.254 - 1))$
 $= 1.817$

Stress factor from Table 4.16.4 [U]:

$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$
 $= (1.254^2(1 + 8.55246 \cdot \log_{10}(1.254)) - 1) / (1.36136(1.254^2 - 1) \cdot (1.254 - 1))$
 $= 9.578$

Stress factor from Table 4.16.4 [Z]:

$= (K^2 + 1) / (K^2 - 1)$
 $= (1.254^2 + 1) / (1.254^2 - 1)$
 $= 4.497$

Stress factor from Table 4.16.4 for integral flanges [d]:

$= U \cdot g_o^2 \cdot h_o / V$
 $= 9.578 \cdot 13.0^2 \cdot 93.048 / 0.371$
 $= 406256.875 \text{ mm}^3$

Stress factor from Table 4.16.4 for integral flanges [e]:

$= F / h_o$
 $= 0.879 / 93.048$
 $= 0.009 \text{ 1/mm}$

Stress factor from Table 4.16.4 [L]:

$= (t \cdot e + 1) / T + t^3/d$
 $= (70.0 \cdot 0.009 + 1) / 1.817 + 70.0^3 / 406256.875$
 $= 1.759$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$
 $= (1.0472 \cdot 7052 / 666.0) / (1.7586 \cdot 19.0^2)$
 $= 171.28 \text{ N./mm}^2$

Longitudinal Hub Stress, Seating [SHa]:

$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$
 $= (1.0472 \cdot 6306 / 666.0) / (1.7586 \cdot 19.0^2)$
 $= 153.16 \text{ N./mm}^2$

Radial Flange Stress, Operating [SRo]:

$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$
 $= (1.8816 \cdot 7052 / 666.0) / (1.7586 \cdot 70.0^2)$
 $= 22.67 \text{ N./mm}^2$

Radial Flange Stress, Seating [SRa]:

$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$
 $= (1.8816 \cdot 6306 / 666.0) / (1.7586 \cdot 70.0^2)$
 $= 20.27 \text{ N./mm}^2$



Tangential Flange Stress, Operating [STo]:
 $= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRO$
 $= (8.7159 \cdot 7052 / (70.0^2 \cdot 666.0)) - 4.4971 \cdot 23$
 $= 82.73 \text{ N./mm}^2$

Tangential Flange Stress, Seating [STa]:
 $= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRA$
 $= (8.7159 \cdot 6306 / (70.0^2 \cdot 666.0)) - 4.4971 \cdot 20$
 $= 73.98 \text{ N./mm}^2$

Average Flange Stress, Operating [SAo]:
 $= (SHo + \max(SRo, STo)) / 2$
 $= (171 + \max(23, 83)) / 2$
 $= 127.01 \text{ N./mm}^2$

Average Flange Stress, Seating [SAa]:
 $= (SHa + \max(SRa, STa)) / 2$
 $= (153 + \max(20, 74)) / 2$
 $= 113.57 \text{ N./mm}^2$

Bolt Stress, Operating [BSo]:
 $= Wm1 / Ab$
 $= 168382 / 100.0794$
 $= 165.00 \text{ N./mm}^2$

Bolt Stress, Seating [BSa]:
 $= (Wm2 / Ab)$
 $= (126327 / 100.0794)$
 $= 123.79 \text{ N./mm}^2$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	171.28	207.00	153.16 207.00
Radial Flange	22.67	138.00	20.27 138.00
Tangential Flange	82.73	138.00	73.98 138.00
Maximum Average	127.01	138.00	113.57 138.00
Bolting	165.00	172.00	123.79 172.00

Minimum Required Flange Thickness 65.481 mm.

Estimated M.A.W.P. (Operating) 33 bars
 Estimated Finished Weight of Flange at given Thk. 120.9 kg.
 Estimated Unfinished Weight of Forging at given Thk 159.3 kg.

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 6305.7 / 1.0 \cdot 9806.637 \cdot 0.371 / (1.581 \cdot 202713 \cdot 13.0^2 \cdot 93.048 \cdot 0.3)$
 $= 0.790 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 7051.8 / 1.0 \cdot 9806.637 \cdot 0.371 / (1.581 \cdot 196922 \cdot 13.0^2 \cdot 93.048 \cdot 0.3)$
 $= 0.910 \text{ (should be } \leq 1)$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 6305.7 / 1.0 \cdot 9806.637 \cdot 0.371 / (1.759 \cdot 202713 \cdot 13.0^2 \cdot 93.048 \cdot 0.3)$
 $= 0.710 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 7051.8 / 1.0 \cdot 9806.637 \cdot 0.371 / (1.759 \cdot 196922 \cdot 13.0^2 \cdot 93.048 \cdot 0.3)$



= 0.818 (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.618, Temperature Reduction per Fig. UCS 66.1 = 21 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -35 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	153.25	207.00	134.20	207.00
Radial Flange	27.17	138.00	23.79	138.00
Tangential Flange	65.55	138.00	57.40	138.00
Maximum Average	109.40	138.00	95.80	138.00
Bolting	165.00	172.00	123.79	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 54.305 mm.

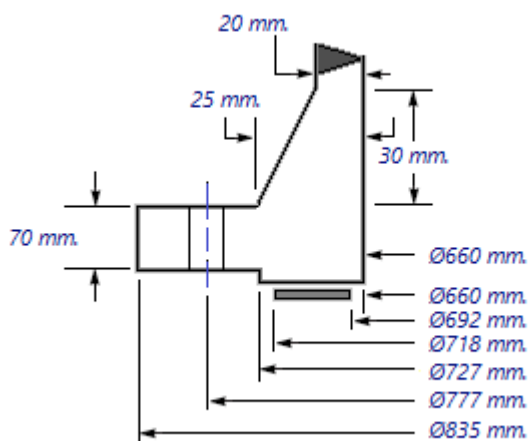
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Flange Input Data Values

Description: SH. FLANGE :

SH. FLANGE

Flange Type	Integral Weld Neck		
Design Pressure	P	32.56	bars
Design Temperature		145	°C
Internal Corrosion Allowance	ci	2.0000	mm.
External Corrosion Allowance	ce	0.0000	mm.
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	660.000	mm.
Flange Outside Diameter	A	835.000	mm.
Flange Thickness	t	70.0000	mm.
Thickness of Hub at Small End	go	20.0000	mm.
Thickness of Hub at Large End	gl	25.0000	mm.
Length of Hub	h	30.0000	mm.
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	777.000	mm.
Nominal Bolt Diameter	a	24.0000	mm.
Type of Threads		TEMA Metric	
Number of Bolts		32	
Flange Face Outside Diameter	Fod	727.000	mm.
Flange Face Inside Diameter	Fid	660.000	mm.
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	718.000	mm.
Gasket Inside Diameter	Gi	692.000	mm.
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	4.5000	mm.
Raised face depth	Fd	0.0000	mm.
Length of Partition Gasket	lp	646.0000	mm.
Width of Partition Gasket	tp	13.0000	mm.
Partition Gasket Factor	mPart	3.7500	
Partition Gasket Design Seating Stress	yPart	62.06	N./mm ²



ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (32.56 \cdot (660.0/2 + 2.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 32.56) + Ca$$

$$= 9.9475 \text{ mm.}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 18.0) / (332.0 + 0.6 \cdot 18.0)$$

$$= 72.457 \text{ bars}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 20.0) / (330.0 + 0.6 \cdot 20.0)$$

$$= 80.696 \text{ bars}$$

Corroded Flange ID,	Bcor = B+2*ci	664.000 mm.
Corroded Large Hub,	g1Cor = g1-ci	23.000 mm.
Corroded Small Hub,	g0Cor = go-ci	18.000 mm.
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	33.500 mm.
Gasket Contact Width,	N = (Go - Gi)/2	13.000 mm.
Basic Gasket Width,	bo = N/2	6.500 mm.
Effective Gasket Width,	b = Cb*sqrt(bo)	6.425 mm.
Gasket Reaction Diameter,	G = Go - 2 * b	705.151 mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 705.1509^2 \cdot 32.565$$

$$= 129687.344 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot lp \cdot bPart \cdot mPart \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 705.1509 \cdot 3.0 \cdot 32.56$$

$$+ 2 \cdot 646.0 \cdot 6.4246 \cdot 3.75 \cdot 32.5647$$

$$= 38694.336 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 664.0^2 \cdot 32.5647$$

$$= 114992.523 \text{ Kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 129687 - 114993$$

$$= 14694.818 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(129687 + 38694 + 0, 0)$$

$$= 168381.672 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.95 \cdot 6.4246 \cdot \pi \cdot 705.151 + 62.06 \cdot 6.4246 \cdot 646.0$$

$$= 126326.547 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((168382 + 0 + 4 \cdot 0 / 705.15) / 172, 126327 / 172)$$

$$= 96.007 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 24.0 + 6 \cdot 70.0 / (3.0 + 0.5)$$

$$= 168.000 \text{ mm.}$$



Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 777.0 \cdot \sin(\pi / 32)$$

$$= 76.159 \text{ mm.}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs / (2a + t)}, 1)$$

$$= \max(\sqrt{76.159 / (2 \cdot 24.0 + 70.0)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	96.007	100.079	
Radial Distance between Hub and Bolts:	33.500	33.500	
Radial Distance between Bolts and Edge:	29.000	29.000	
Circ. Spacing between the Bolts:	57.500	76.159	168.000

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 100.079 \cdot 172.0 / (68.95 \cdot \pi(718.0 + 692.0))$$

$$= 5.636 \text{ mm.}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 16 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 100.0794$$

$$= 175523.97 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 168382 - 129687$$

$$= 38694.33 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (777.0 - 705.1509) / 2$$

$$= 35.9245 \text{ mm.}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((777.0 - 664.0) / 2 + 35.9245)$$

$$= 46.2123 \text{ mm.}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (777.0 - 664.0 - 23.0) / 2$$

$$= 45.0000 \text{ mm.}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	114993.	45.0000	1.0000	5175.
Face Pressure, Mt	14695.	46.2123	1.0000	679.
Gasket Load, Mg	38694.	35.9245	1.0000	1390.
Gasket Seating, Matm	175524.	35.9245	1.0000	6306.

Total Moment for Operation, Mo 7244. Kg-m.
 Total Moment for Gasket seating, Mg 6306. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 109.325 mm.
 Hub Ratio, h/h0 = HL / H0 0.274



Thickness Ratio, $g1/g0 = (g1Cor/goCor)$ 1.278

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.896
 Factor V per 4.16.5 0.431
 Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

= A / B
 = 835.0/664.0
 = 1.258

Stress factor from Table 4.16.4 [Y]:

= $(0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$
 = $(0.66845 + (5.7169(1.258^2 \cdot \log_{10}(1.258))) / (1.258^2 - 1)) / (1.258 - 1)$
 = 8.605

Stress factor from Table 4.16.4 [T]:

= $(K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 = $(1.258^2(1 + 8.55246 \cdot \log_{10}(1.258)) - 1) / ((1.0472 + 1.9448 \cdot 1.258^2) \cdot (1.258 - 1))$
 = 1.815

Stress factor from Table 4.16.4 [U]:

= $(K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$
 = $(1.258^2(1 + 8.55246 \cdot \log_{10}(1.258)) - 1) / (1.36136(1.258^2 - 1) \cdot (1.258 - 1))$
 = 9.456

Stress factor from Table 4.16.4 [Z]:

= $(K^2 + 1) / (K^2 - 1)$
 = $(1.258^2 + 1) / (1.258^2 - 1)$
 = 4.440

Stress factor from Table 4.16.4 for integral flanges [d]:

= $U \cdot g0^2 \cdot h0 / V$
 = $9.456 \cdot 18.0^2 \cdot 109.325 / 0.431$
 = 776736.750 mm.³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
 = 0.896/109.325
 = 0.008 1/mm.

Stress factor from Table 4.16.4 [L]:

= $(t \cdot e + 1) / T + t^3/d$
 = $(70.0 \cdot 0.008 + 1) / 1.815 + 70.0^3/776737$
 = 1.308

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= $(f \cdot Mop / Bcor) / (L \cdot g1^2)$
 = $(1.0 \cdot 7244/664.0) / (1.3084 \cdot 23.0^2)$
 = 154.57 N./mm²

Longitudinal Hub Stress, Seating [SHa]:

= $(f \cdot Matm / Bcor) / (L \cdot g1^2)$
 = $(1.0 \cdot 6306/664.0) / (1.3084 \cdot 23.0^2)$
 = 134.55 N./mm²

Radial Flange Stress, Operating [SRo]:

= $(\beta \cdot Mop / Bcor) / (L \cdot t^2)$
 = $(1.7647 \cdot 7244/664.0) / (1.3084 \cdot 70.0^2)$
 = 29.45 N./mm²

Radial Flange Stress, Seating [SRa]:

= $(\beta \cdot Matm / Bcor) / (L \cdot t^2)$
 = $(1.7647 \cdot 6306/664.0) / (1.3084 \cdot 70.0^2)$
 = 25.63 N./mm²

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRO$$

$$= (8.6047 \cdot 7244 / (70.0^2 \cdot 664.0)) - 4.4401 \cdot 29$$

$$= 57.12 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRA$$

$$= (8.6047 \cdot 6306 / (70.0^2 \cdot 664.0)) - 4.4401 \cdot 26$$

$$= 49.72 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (155 + \max(29, 57)) / 2$$

$$= 105.85 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHA + \max(SRA, STA)) / 2$$

$$= (135 + \max(26, 50)) / 2$$

$$= 92.14 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 168382 / 100.0794$$

$$= 165.00 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (126327 / 100.0794)$$

$$= 123.79 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	154.57	207.00	134.55	207.00
Radial Flange	29.45	138.00	25.63	138.00
Tangential Flange	57.12	138.00	49.72	138.00
Maximum Average	105.85	138.00	92.14	138.00
Bolting	165.00	172.00	123.79	172.00

Minimum Required Flange Thickness 52.705 mm.

Estimated M.A.W.P. (Operating) 34 bars
 Estimated Finished Weight of Flange at given Thk. 122.7 kg.
 Estimated Unfinished Weight of Forging at given Thk 159.3 kg.

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 6305.7 / 1.0 \cdot 9806.637 \cdot 0.431 / (0.977 \cdot 202713 \cdot 18.0^2 \cdot 109.325 \cdot 0.3)$$

$$= 0.660 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 7244.0 / 1.0 \cdot 9806.637 \cdot 0.431 / (0.977 \cdot 195371 \cdot 18.0^2 \cdot 109.325 \cdot 0.3)$$

$$= 0.787 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 6305.7 / 1.0 \cdot 9806.637 \cdot 0.431 / (1.308 \cdot 202713 \cdot 18.0^2 \cdot 109.325 \cdot 0.3)$$

$$= 0.493 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 7244.0 / 1.0 \cdot 9806.637 \cdot 0.431 / (1.308 \cdot 195371 \cdot 18.0^2 \cdot 109.325 \cdot 0.3)$$

$$= 0.588 \text{ (should be } \leq 1)$$



Minimum Design Metal Temperature Results:

Thickness Ratio = 0.468, Temperature Reduction per Fig. UCS 66.1 = 37 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -44 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	144.59	207.00	123.62	207.00
Radial Flange	31.91	138.00	27.29	138.00
Tangential Flange	49.45	138.00	42.28	138.00
Maximum Average	97.02	138.00	82.95	138.00
Bolting	165.00	172.00	123.79	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 48.971 mm.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
CH. COVER		25.065	18	3	660	138	1
CH. SHELL		25.065	16	3	660	138	2
CH. FLANGE		25.065	70	3	660	138	3
SH. FLANGE		32.565	70	2	660	138	4
MAIN SHELL		32.565	20	2	660	138	5
SH. COVER		32.565	23	2	660	138	6

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
CH. COVER		25	54.2364	66.5812	16	8.98794	1
CH. SHELL		25	52.5718	65.0127	16	9.1154	2
CH. FLANGE		25	32.5098	32.5745	70	65.4812	3
SH. FLANGE		32.5	33.8813	33.946	70	52.705	4
MAIN SHELL		32.5	72.3917	80.6956	20	9.94758	5
SH. COVER		32.5	74.9391	83.1262	20	9.79107	6

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

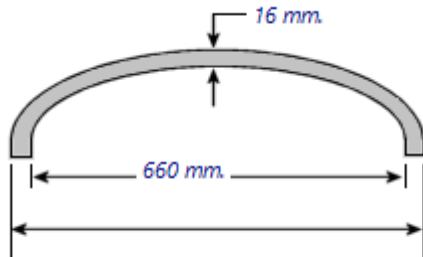
Shell Side MAWP = 34 bars
 Shell Side MAPnc = 34 bars
 Channel Side MAWP = 33 bars
 Channel Side MAPnc = 33 bars

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: CH. COVER

Elliptical Head Design Information:							
Design Pressure	P	25.065	bars	Int. Design Temperature		120	°C
Material		SA-516	70	Ext. Design Temperature		120	°C
Length		50.000	mm.	External Pressure Chart		CS-2	
				UNS Number		K02700	
Int. Corr. All.	c	3.0000	mm.	Allowable Stress (ope)	S	138.0	N./mm ²
Ext. Corr. All.	ce	0.0000	mm.	Allowable Stress (amb)	Sa	138.0	N./mm ²
Inside Diameter	Di	660.000	mm.	Actual Stress		63.7	N./mm ²
Spec. Min. Thk t _e , or s _a		16.0000	mm.	Spec. Nominal Thk.		18.000	mm.
Surface Area		6344.3	cm ²	SG of Contents		1.000	
Element Volume		54748.9	cm ³	Weight of Contents		54.7	Kgf
Empty Weight		91.6	Kgf	Operating Weight		146.3	Kgf
				Comp. All. Ope. Stress		142.4	N./mm ²
				Aspect Ratio		2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 8.250 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 4.125 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (25.065 \cdot 666.0 \cdot 0.988) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 25.065)$$

$$= 5.9879 + 3.0000 = 8.9879 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.065 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 13.0) / (0.988 \cdot 666.0 + 0.2 \cdot 13.0)$$

$$= 54 - 0.1 = 54 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 16.0) / (1.0 \cdot 660.0 + 0.2 \cdot 16.0)$$

$$= 67 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (25.065 \cdot (0.988 \cdot 666.0 + 0.2 \cdot 13.0)) / (2 \cdot 1.0 \cdot 13.0)$$

$$= 63.698 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (25.065 \cdot 333.0) / (138.0 \cdot 1.0 - 0.6 \cdot 25.065) + 3.0$$

$$= 9.115 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 0.065 bars](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 15.0) / (333.0 + 0.6 \cdot 15.0)$$

$$= 61 - 0.1 = 60 \text{ bars}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (666.0 / (2 \cdot 168.0))^2) / 6$$

$$= 0.988148$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 11.139 %

[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 16.0$, $t_r = 7.771$, $c = 3.0$ mm., $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.598$, Temp. Reduction = 23 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-37 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 18.0$, $t_r = 7.958$, $c = 3.0$ mm., $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.531$, Temp. Reduction = 29 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-39 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	32.510 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	18.000 mm.	Basic MDMT	-11 °C
Stress/Thk. Ratio	0.531	Temperature Difference	29 °C
UCS-66(c) red.	No	Computed Min. Temp.	-39 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

[Note: Temperature values displayed above are for the location with the highest basic MDMT](#)

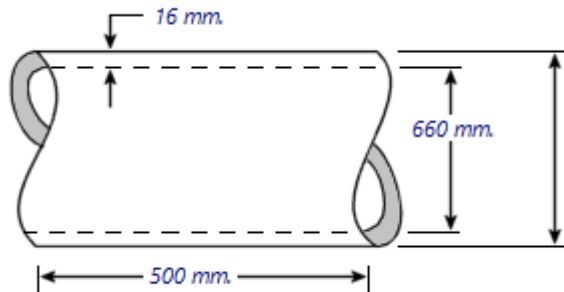
Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 8.250 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 4.125 mm.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: CH. SHELL

Cylindrical Shell Design Information:					
Design Pressure	P	25.065 bars	Int. Design Temperature		120 °C
Material		SA-516 70	Ext. Design Temperature		120 °C
Length		500.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	660.000 mm.	Actual Stress		65.7 N./mm ²
Spec. Min. Thk t _e , or s _a		16.0000 mm.	Spec. Nominal Thk.		16.000 mm.
Surface Area		10869.9 cm ²	SG of Contents		1.000
Element Volume		171090.4 cm ³	Weight of Contents		171.0 Kg
Empty Weight		131.7 Kg	Operating Weight		302.7 Kg
			Comp. All. Ope. Stress		142.4 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (25.065 \cdot 333.0) / (138.0 \cdot 1.0 - 0.6 \cdot 25.065)$$

$$= 6.1154 + 3.0000 = 9.1154 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.065 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 13.0) / (333.0 + 0.6 \cdot 13.0)$$

$$= 53 - 0.1 = 53 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 16.0) / (330.0 + 0.6 \cdot 16.0)$$

$$= 65 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (25.065 \cdot (333.0 + 0.6 \cdot 13.0)) / (1.0 \cdot 13.0)$$

$$= 65.713 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 2.367 %

Minimum Design Metal Temperature Results:

Govern. thk, t_g = 16.0, t_r = 7.958, c = 3.0 mm., E* = 1.0



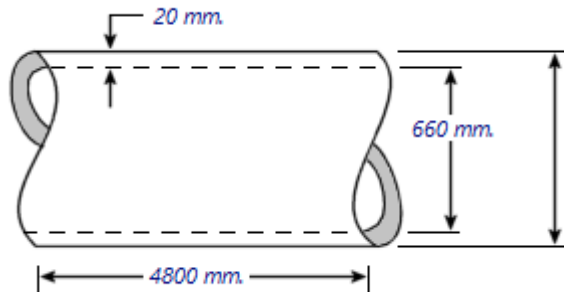
Thickness Ratio = $tr * E^* / (tg - c) = 0.612$, Temp. Reduction = 22 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -36 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	32.510 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	16.000 mm.	Basic MDMT	-14 °C
Stress/Thk. Ratio	0.612	Temperature Difference	22 °C
UCS-66(c) red.	No	Computed Min. Temp.	-36 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: MAIN SHELL

Cylindrical Shell Design Information:					
Design Pressure	P	32.565 bars	Int. Design Temperature		145 °C
Material		SA-516 70	Ext. Design Temperature		120 °C
Length		4800.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	660.000 mm.	Actual Stress		62.0 N./mm ²
Spec. Min. Thk t,e, or sa		20.0000 mm.	Spec. Nominal Thk.		20.000 mm.
Surface Area		105557.5 cm ²	SG of Contents		1.000
Element Volume		1018460.7 cm ³	Weight of Contents		1017.8 Kg
Empty Weight		1589.5 Kg	Operating Weight		2607.3 Kg
			Comp. All. Ope. Stress		146.3 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (32.565 \cdot 332.0) / (138.0 \cdot 1.0 - 0.6 \cdot 32.565)$$

$$= 7.9476 + 2.0000 = 9.9476 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.065 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 18.0) / (332.0 + 0.6 \cdot 18.0)$$

$$= 72 - 0.1 = 72 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 20.0) / (330.0 + 0.6 \cdot 20.0)$$

$$= 81 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (32.565 \cdot (332.0 + 0.6 \cdot 18.0)) / (1.0 \cdot 18.0)$$

$$= 62.022 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 2.941 %

Minimum Design Metal Temperature Results:

Govern. thk, tg = 20.0, tr = 8.274, c = 2.0 mm., E* = 1.0



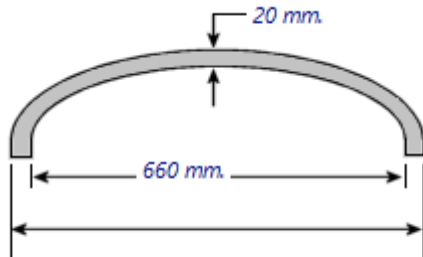
Thickness Ratio = $tr * E^* / (tg - c) = 0.46$, Temp. Reduction = 38 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -46 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	33.881 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	20.000 mm.	Basic MDMT	-7 °C
Stress/Thk. Ratio	0.460	Temperature Difference	38 °C
UCS-66(c) red.	No	Computed Min. Temp.	-46 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: SH. COVER

Elliptical Head Design Information:					
Design Pressure	P	32.565 bars	Int. Design Temperature		145 °C
Material		SA-516 70	Ext. Design Temperature		120 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	660.000 mm.	Actual Stress		59.9 N./mm ²
Spec. Min. Thk t _e , or s _a		20.0000 mm.	Spec. Nominal Thk.		23.000 mm.
Surface Area		6512.0 cm ²	SG of Contents		1.000
Element Volume		54748.9 cm ³	Weight of Contents		54.7 Kg _f
Empty Weight		118.7 Kg _f	Operating Weight		173.4 Kg _f
			Comp. All. Ope. Stress		146.3 N./mm ²
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 8.250 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 4.125 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (32.565 \cdot 664.0 \cdot 0.992) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 32.565)$$

$$= 7.7911 + 2.0000 = 9.7911 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.065 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 18.0) / (0.992 \cdot 664.0 + 0.2 \cdot 18.0)$$

$$= 75 - 0.1 = 75 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 20.0) / (1.0 \cdot 660.0 + 0.2 \cdot 20.0)$$

$$= 83 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (32.565 \cdot (0.992 \cdot 664.0 + 0.2 \cdot 18.0)) / (2 \cdot 1.0 \cdot 18.0)$$

$$= 59.915 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$\begin{aligned} &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)} \\ &= (32.565 \cdot 332.0) / (138.0 \cdot 1.0 - 0.6 \cdot 32.565) + 2.0 \\ &= 9.948 \text{ mm.} \end{aligned}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.065 bars

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)} \\ &= (138.0 \cdot 1.0 \cdot 21.0) / (332.0 + 0.6 \cdot 21.0) \\ &= 84 - 0.1 = 84 \text{ bars} \end{aligned}$$

Factor K, corroded condition [Kcor]:

$$\begin{aligned} &= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6 \\ &= (2 + (664.0 / (2 \cdot 167.0))^2) / 6 \\ &= 0.992040 \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 13.945 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 20.0$, $t_r = 8.107$, $c = 2.0$ mm., $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.45$, Temp. Reduction = 40 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-47 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 23.0$, $t_r = 8.274$, $c = 2.0$ mm., $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.394$, Temp. Reduction = 52 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	33.881 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	23.000 mm.	Basic MDMT	-3 °C
Stress/Thk. Ratio	0.394	Temperature Difference	45 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 8.250 mm.
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 4.125 mm.

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	44.046 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	42.250 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	44.065 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	37.269 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	46.475 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	46.475 bars
App 27-4	= M.A.W.P.	33.881 bars

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	42.263 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	32.500 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	42.282 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	35.761 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	35.750 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	35.750 bars
App 27-4	= M.A.W.P.	32.510 bars

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
CH. COVER	107.6	234.0	0.460	42.33
CH. SHELL	111.0	234.0	0.474	42.33
MAIN SHELL	84.0	179.4	0.468	44.11
SH. COVER	81.2	234.0	0.347	44.11

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
TS IN NOZ. T1	Nozzle	138.00	138.00	1.000
TS OUT NOZ. T2	Nozzle	138.00	138.00	1.000
SS IN NOZ. S1	Nozzle	138.00	138.00	1.000
SS OUT NOZ. S2	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
CH. COVER	138.00	138.00	1.000
CH. SHELL	138.00	138.00	1.000
CH. FLANGE	138.00	138.00	1.000
SH. FLANGE	138.00	138.00	1.000
MAIN SHELL	138.00	138.00	1.000
SH. COVER	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	92.40	92.40	1.000
Tubesheet Material	138.00	138.00	1.000



Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
TS IN NOZ. T1	48.19	179.40	0.269
TS OUT NOZ. T2	48.19	179.40	0.269
SS IN NOZ. S1	87.24	179.40	0.486
SS OUT NOZ. S2	87.24	179.40	0.486

Elements Suitable for Test Pressure.

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Input Echo, Tubesheet Number 1, Description: Tubesheet

Shell Data:

Main Shell Description: MAIN SHELL

Shell Test Pressure		44.05 bars
Shell Thickness	ts	20.0000 mm.
Shell Internal Corrosion Allowance	cas	0.0000 mm.
Shell External Corrosion Allowance	casext	0.0000 mm.
Inside Diameter of Shell	Ds	660.000 mm.
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: CH. SHELL

Channel Type:		Cylinder
Channel Test Pressure		42.26 bars
Channel Thickness	tc	16.0000 mm.
Channel Internal Corrosion Allowance	cac	0.0000 mm.
Channel External Corrosion Allowance	cacext	0.0000 mm.
Inside Diameter of Channel	Dc	660.000 mm.
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	442
Tube Wall Thickness	et	2.1100 mm.
Tube Outside Diameter	D	19.0500 mm.
Total Straight Tube Length	Lt	4600.00 mm.
Straight Tube Length (bet. inner tubsht faces) L		4515.00 mm.
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		92.40 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	180.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.4000 mm.
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm.
Groove Weld Leg	ag	2.1100 mm.
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	313.500 mm.
Largest Center-to-Center Tube Distance	Ul	57.1500 mm.
Length of Expanded Portion of Tube	ltx	0.0000 mm.
Tube-side pass partition groove depth	hg	6.0000 mm.

Tubesheet Data:

Tubesheet TYPE: U-tube, Gasketed both Sides, Conf. d

Tubesheet Design Metal Temperature	T	21.11 °C
Tubesheet Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	85.0000 mm.
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm.
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm.
Tubesheet Outside Diameter	A	835.000 mm.

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		No
Step 1 Diameter on the Tubeside	dt1	724.00 mm.
Step 1 Depth on the Tubeside	ht1	6.00 mm.
Step 2 Diameter on the Tubeside	dt2	657.00 mm.
Step 2 Depth on the Tubeside	ht2	6.00 mm.
Is the Tubesheet Flat on Shellside?		No
Step 1 Diameter on the Shellside	ds1	724.00 mm.
Step 1 Depth on the Shellside	hs1	6.00 mm.
Step 2 Diameter on the Shellside	ds2	657.00 mm.
Step 2 Depth on the Shellside	hs2	6.00 mm.
Calculated Tubesheet Diameter per Div 2, 4.18.4		724.00 mm.

Note: Tubesheet diameter is now: 724.000 mm. Div 2, 4.18.4.

Dimension G for the Channel Side	Gc	705.151 mm.
Area of the Untubed Lanes	AL	369.2 cm ²

Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Both
Flange Outside Diameter	A	835.000 mm.
Flange Inside Diameter	B	660.000 mm.
Flange Face Outside Diameter	Fod	727.000 mm.
Flange Face Inside Diameter	Fid	660.000 mm.
Gasket Outside Diameter	Go	718.000 mm.
Gasket Inside Diameter	Gi	692.000 mm.
Small end Hub thk.	g0	20.0000 mm.
Large end Hub thk.	g1	25.0000 mm.
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	68.95 N./mm ²
Flange Facing Sketch	Code Sketch 1a	
Gasket Thickness	tg	4.5000 mm.
Full face Gasket Flange Option	Program Selects	
Length of Partition Gasket	lp	646.000 mm.
Width of Partition Gasket	wp	13.0000 mm.
Partition Gasket Factor,	mPart	3.7500
Partition Gasket Design Seating Stress	yPart	62.06 N./mm ²
Partition Gasket Facing Sketch	Code Sketch 1a	
Partition Gasket Column for Gasket Seating	Code Column II	

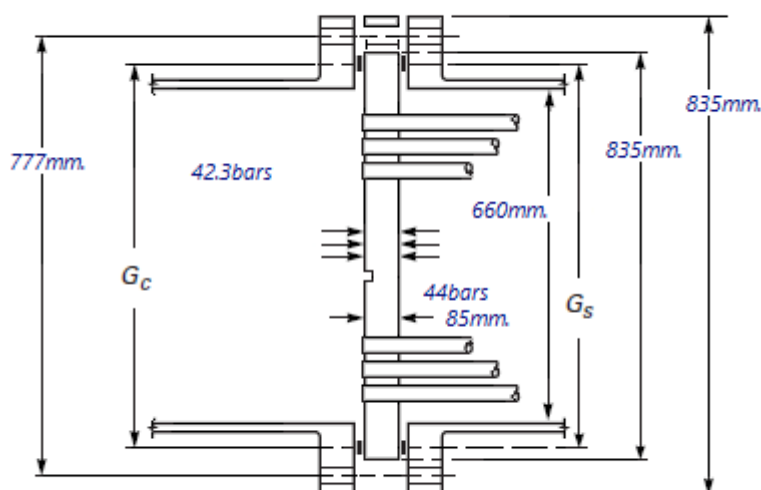
Bolting Information:

Diameter of Bolt Circle	C	777.000 mm.
Nominal Bolt Diameter	dB	24.0000 mm.
Thread Series	TEMA Metric Thread	
Number of Bolts	n	32
Bolt Material	SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm.

Alternate Flange Operating Bolt Load, Wm1	227729.78 Kg
Alternate Flange Seating Bolt Load, Wm2	126326.55 Kg
Alternate Flange Design Bolt Load, W	201626.88 Kg

Tubesheet Integral with	None
Tubesheet Extended as Flange	Yes
Thickness of Extended Portion of Tubesheet	Tf 61.0000 mm.
Is Bolt Load Transferred to the Tubesheet	Yes

Tubesheet Gasketed w/Shell and Channel



Configuration d:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (42.26 \cdot (19.05/2 - 0)) / (92.4 \cdot 1.0 + 0.4 \cdot 42.26)$
 $= 0.4279 + 0.0000 = 0.4279 \text{ mm.}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 21.11 °C
 Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):



TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	4828.50	9.03	50.0000	0.0134948	95.84

EMAWP = $(2.167 / (D/T) - 0.0833) * B = 150.192$ bars

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.7407	19.05	4828.50	25.72	50.0000	0.0016630	84.97

EMAWP = $(4*B) / (3*(D/T)) = (4 * 84.975) / (3 * 25.7186) = 44.0512$ bars

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.7407 mm.
Allowable Internal Pressure at Corroded thickness	224.57 bars
Required Internal Design Pressure	42.26 bars
Allowable External Pressure at Corroded thickness	150.19 bars
Required External Design Pressure	44.05 bars
Required Thickness due to Shell Side pressure	0.7407 mm.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (G_{oc} - G_{ic}) / 2$	13.000 mm.
Basic Gasket Width,	$b_0 = N / 2$	6.500 mm.
Effective Gasket Width,	$b = \sqrt{b_0} * 2.5$	6.422 mm.
Gasket Reaction Diameter,	$G = G_o - 2.0 * b$	705.156 mm.

Warning: Available bolt area, 100.079 cm² for then shell flange is < required 129.846 cm²

Note that for the pressure test case, ASME VIII-2 paragraph 4.18.13 (c)(9) does not require that the bolt area requirement be met.

Results for ASME U-tube Tubesheet Calculations for Configuration d:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$= 2 * r_o + dt$
 $= 2 * 313.5 + 19.05$
 $= 646.050$ mm.

Basic ligament efficiency for shear [μ]:

$= (p - dt) / p$
 $= (25.4 - 19.05) / 25.4$
 $= 0.250$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ρ_s]:

$= G_s / D_o$ (Configurations d, e, f)
 $= 705.1556 / 646.05$
 $= 1.0915$

Ratio [ρ_c]:

$= G_c / D_o$ (Configurations d)
 $= 705.1509 / 646.05$
 $= 1.0915$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$= l_{tx} / h$
 $= 0 / 85.0$
 $= 0.000$ (must be $0 \leq \rho \leq 1$)

Effective tube hole Diameter [d*]:
 $= \max(dt - 2tt(EtT/E) \cdot (StT/S) \cdot (p) ; dt - 2tt)$
 $= \max(19.05 - 2 \cdot 2.11 (0.20270E+09 / 2.0270E+09) \cdot$
 $(92/137) \cdot (0) ; 19.05 - 2 \cdot 2.11)$
 $= 19.0500 \text{ mm.}$

Effective tube pitch [p*]:
 $= p / \sqrt{ (1 - 4 \cdot \min(A1 ; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$
 $= 25.4 / \sqrt{ (1 - 4 \cdot \min(369.19 ; 4 \cdot 646.05 \cdot 25.4) / (\pi \cdot 646.05^2)) }$
 $= 26.9637 \text{ mm.}$

Effective ligament efficiency for bending [μ^*]:
 $= (p^* - d^*) / p^*$
 $= (26.9637 - 19.05) / 26.9637$
 $= 0.2935$

$h/p = 3.346457$; $\mu^* = 0.293495$; $E^* = 55264016 \text{ KPa.}$
 $E^*/E = 0.272638$; $\nu^* = 0.358419$

Note: As h/p (3.346) is > 2 , data values for $h/p = 2$ were used.

Skip Step 4 for Configuration d :

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:
 $= A / Do$
 $= 724.0 / 646.05$
 $= 1.1207$

Coefficient [F]:
 $= (1 - \nu^*) / E^* \cdot (E \cdot \ln(K))$
 $= (1 - 0.36) / 55264016 \cdot (0.20270E+09 \cdot \ln(1.12))$
 $= 0.2681$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:
 $= Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt]$
 $= 646.05^2/16 \cdot [(1.091 - 1) \cdot (1.091^2 + 1) \cdot 44.046 -$
 $(1.091 - 1) \cdot (1.091^2 + 1) \cdot 42.263]$
 $= 9343.5781 \text{ bars} \cdot \text{mm.}^2$

Moment acting on unperforated tubesheet rim [M*]:
 $= Mts + W^* \cdot (Gc - Gs) / (2 \cdot \pi \cdot Do)$
 $= 9343.6 + 227729.8 \cdot (705.151 - 705.151) / (2 \cdot \pi \cdot 646.05)$
 $= 9343.5781 \text{ bars} \cdot \text{mm.}^2$

Note: W^* is the maximum of the bolt loads between the shell and channel sides.

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:
 $= ((M^*) - Do^2/32 \cdot F(Ps - Pt)) / (1 + F)$
 $= ((9343.58) - 646.05^2/32 \cdot 0.268 (44.05 - 42.26)) / (1 + 0.27)$
 $= 2452.2549 \text{ bars} \cdot \text{mm.}^2$

Maximum bending moment acting on center of tubesheet [Mo]:
 $= Mp + Do^2/64 \cdot (3 + \nu^*) \cdot (Ps - Pt)$
 $= 2452.25 + 646.05^2/64 \cdot (3 + 0.358) \cdot (44.05 - 42.26)$
 $= 41502.5664 \text{ bars} \cdot \text{mm.}^2$

Maximum bending moment acting on tubesheet [M]:
 $= \max(| Mp | ; | Mo |)$
 $= \max(| 2452.255 | ; | 41502.566 |)$
 $= 41502.5664 \text{ bars} \cdot \text{mm.}^2$

ASME Section VIII-2 Part 4.18.7.4 Step 8:



Tubesheet Bending Stress at original thickness:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 41502.566 / (0.2935 (85.0 - 6.0)^2) \\ &= 13.5955 \text{ N./mm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned} &= 2 \cdot \beta T \cdot S_y \\ &= 2 \cdot 0.95 \cdot 250.0 \\ &= 474.99 \text{ N./mm}^2 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 40992.664 / (0.2935 (19.3022 - 6.0)^2) \\ &= 473.6230 \text{ N./mm}^2 \end{aligned}$$

Required tubesheet thickness for Bending Stress [H_{reqB}]:

$$\begin{aligned} &= h + C_{ats} + C_{atc} \\ &= 19.3022 + 0 + 0 \\ &= 19.3022 \text{ mm.} \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [H_{req}]:

$$\begin{aligned} &= \max(H_{reqB}; H_{reqS}) \\ &= \max(19.3022; 0.6063) \\ &= 19.3022 \text{ mm.} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$\begin{aligned} &= |P_s - P_t| \\ &= |44.05 - 42.26| \\ &= 1.783 \text{ bars} \end{aligned}$$

Shear Stress check [τ_{limit}]:

$$\begin{aligned} &= 4 \cdot \mu \cdot h / D_o \cdot 0.8 \cdot \beta T \cdot S_y \\ &= 4 \cdot 0.25 \cdot 85.0 / 646.05 \cdot 0.8 \cdot 0.95 \cdot 249.995 \\ &= 24.998 \text{ N./mm}^2 \end{aligned}$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$\begin{aligned} &= 1 / (4 \cdot \mu) \cdot (D_o / (0.8 \cdot \beta T \cdot S_y)) \cdot |P_s - P_t| + C_{ats} + C_{atc} \\ &= 1 / (4 \cdot 0.25) \cdot (646.05 / 189.997) \cdot |44.05 - 42.26| + 0 \\ &= 0.6063 \text{ mm.} \end{aligned}$$

Note: Analysis Completed for Tubesheet Configuration d.

Tube Metal Area [A_t]:

$$\begin{aligned} &= \pi (d_o - t) t \\ &= \pi (19.05 - 2.11) 2.11 \\ &= 1.123 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per UW-20.3:

Tube Strength [F_t]:

$$\begin{aligned} &= \pi \cdot t (d_o - t) S \\ &= \pi \cdot 2.11 (19.05 - 2.11) 170.997 \\ &= 1957.967 \text{ Kgf} \end{aligned}$$

Groove Weld Strength [F_g]:

$$\begin{aligned} &= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t) \\ &= 0.85 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 171.0 \\ &= 1957.9670 \text{ Kgf} \end{aligned}$$

Weld Strength Factor [f_w]:

$$\begin{aligned} &= S / S_w \text{ where } S_w = \min(S \text{ and } S_t) \\ &= 170.997 / 170.997 \\ &= 1.000 \end{aligned}$$

Design Strength Ratio [f_d]:

= 1 for full strength welds

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 171.0$$

$$= 1300.8848 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) f_w \cdot f_d} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 19.05)^2 + 1.07 \cdot 2.11 \cdot (19.05 - 2.11) 1.0 \cdot 1.0} - 0.75 \cdot 19.05 \right)$$

$$= 2.562 \text{ mm.}$$

Minimum Required Fillet Weld Leg [afr]: 1.2810 mm.

Minimum Required Groove Weld Leg [agr]: 1.2810 mm.

Tube-Tubesheet Jt allowable, 1957.97 is >= tube strength 1957.97 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	13.6 <=	475.0 N./mm ²	Ok
Tubesheet shear stress	1.4 <=	190.0 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm.)	Required	Actual	P/F
Tubesheet Thickness :	19.302	85.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.281	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.281	2.110	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	---- Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	71.159 ...	322 475	32 190	Ps+Pt D1	Ok
D2uc	72.506 30.169	336 475	33 190	Ps+Pt D2	Ok
D3uc	19.302 ...	14 475	1 190	Ps+Pt D3	Ok
Max:	72.5057 30.169 mm.	0.707	0.176 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm.)	Required	Actual	P/F
Tubesheet thickness :	72.506	85.000	Ok
Tubesheet thickness flanged extension :	30.169	61.000	Ok
Tube thickness :	0.741	2.110	Ok
Tube-tubesheet fillet weld leg :	1.281	2.110	Ok
Tube-tubesheet groove weld leg :	1.281	2.110	Ok

Note: This is a full strength Tube to Tubesheet Joint.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	No Calc	692	13	0.0026092	109.321
20	30	605	692	13	0.0040704	116.751
30	40	No Calc	...	67	No Calc	No Calc
40	50	No Calc	...	68	No Calc	No Calc
50	60	4905	700	18	0.00080664	78.4191
60	70	No Calc	700	18	0.0035714	114.81

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	16	No Calc	...	22.8178
20	30	16	No Calc	...	29.2422
30	40	70	59.1566	...	No Calc
40	50	70	45.72	...	No Calc
50	60	20	No Calc	...	26.885
60	70	20	No Calc	...	32.8008

Minimum

23

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	605	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	4905	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	91.612	54748.9	76.3433	56097.3	9.1612
20	30	131.679	171090	107.464	174215	13.1679
30	40	120.904	34655.6	116.061	34843.1	12.0904
40	50	122.697	34509.3	119.474	34634.1	12.2697
50	60	1589.5	1018460	1434.75	1038429	158.95
60	70	118.652	54748.9	108.334	55645.4	11.8652
Total		2175	1368213.75	1962	1393864.88	217



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
10	Liqd	54.7155	-55	...	...	Liquid 10
20	Liqd	170.986	250	...	...	Liquid: 20
20	Nozl	82.2397	250	-431.6	...	TS IN NOZ. T1
20	Nozl	82.2397	250	431.6	...	TS OUT NOZ. T2
30	Liqd	34.6344	50	...	...	Liquid 30
40	Liqd	34.4882	50	...	...	Liquid 40
50	Sadl	78.0179	960	524	...	FIXED SADDLE
50	Sadl	78.0179	3840	524	...	RHS Saddle
50	Liqd	1017.84	2400	...	...	Liquid 50
50	Nozl	169.375	400	-533.2	...	SS IN NOZ. S1
50	Nozl	186.358	4300	533.2	...	SS OUT NOZ. S2
60	Liqd	54.7155	105	...	...	Liquid 60
30	FTsh	305.536	145.675	...	...	Tubesheet
30	Tube	2096.13	2403.18	...	...	Tubes
40	Bafl	20.6842	770	...	...	Baffles
40	Bafl	20.6842	1220	...	...	Baffles
40	Bafl	20.6842	1670	...	...	Baffles
40	Bafl	20.6842	2120	...	...	Baffles
40	Bafl	20.6842	2570	...	...	Baffles
40	Bafl	20.6842	3020	...	...	Baffles
40	Bafl	20.6842	3470	...	...	Baffles
40	Bafl	20.6842	3920	...	...	Baffles
40	Bafl	20.7794	4370	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	156.0
Liquid	1367.4
Nozzles	520.2
Exchanger Components	2587.9
Liquid in Tubes	392.1

Sum of the Detail Weights 5023.6 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	2392.5	2392.5	2392.5	2392.5	2392.5	2392.5
Saddles	156.0	156.0	156.0	156.0	156.0	156.0
Nozzles	520.2	520.2	520.2	520.2	520.2	520.2
Exchanger	2587.9	2587.9	2587.9	2587.9	2587.9	2587.9

Ope. Liquid	...	...	...	...	...	1367.4
Tube Ope Lqd	...	...	...	...	...	392.1
Test Liquid	...	1367.4	...	...	...	...
Tube Tst Lqd	...	392.1	...	...	...	...
Totals	5656.7	7416.1	5656.7	5656.7	5656.7	7416.1

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	5656.7 kg.
Shop Test Wt.	- Fabricated weight + water (full)	7416.1 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	5656.7 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	5656.7 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	5656.7 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	7416.1 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	7182.3 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	392292.2 cm ³
Weight of Ope Liq in tubes :	392.1 kg.
Weight of Water in tubes :	392.1 kg.

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	6344.27	5758.57
20	30	10869.9	10367.3
30	40	4554.56	2073.45
40	50	4560.22	2073.45
50	60	10555.8	99525.7
60	70	6511.96	5758.57

Total	138398.5	125557.0 cm ²
Total	149.0	135.1 ft ²

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Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
TS IN NOZ. T1	92.08	102.00	120	600	GR 1.1	...	...	...	...
TS OUT NOZ. T2	92.08	102.00	120	600	GR 1.1	...	...	...	...
SS IN NOZ. S1	45.26	51.00	145	300	GR 1.1	...	...	...	...
SS OUT NOZ. S2	45.26	51.00	145	300	GR 1.1	...	...	...	...

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 45.260 bars
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 51.000 bars

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 92.081 bars
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 102.000 bars

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-1, UG-44(b) for more information.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 Km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	0
Base Elevation	300 mm.
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
= $\min(0.85, 0.925((1 + 1.7 \cdot g_Q \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot g_V \cdot I_{zbar})))$
= $\min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.194 \cdot 0.988) / (1 + 1.7 \cdot 3.4 \cdot 0.194)))$
= $\min(0.85, 0.919)$
= 0.850

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.836
Structure Height to Diameter ratio	8.604

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:
Exposure Category: D from Table 26.11-1
α = 11.5: Zg = 589788 mm.

Effective Height [z]:
= Centroid Height + Vessel Base Elevation
= 750.0 + 300.0 = 1050.0 mm.
= 3.445 ft. Imperial Units

Velocity Pressure coefficient evaluated at height z [Kz]:
Because z (3.445 ft.) < 15 ft.
= $2.41(15 / Z_g)^{2/\alpha}$
= $2.41(15 / 1935.0)^{2/11.5}$
= 1.035

Type of Hill: No Hill

Wind Directionality Factor [Kd]:
= 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:
K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.035 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 111.85^2)$$

$$= 33.1 \text{ psf } [161.849 \text{ Kgs/m}^2]$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 33.149 \cdot 0.85 \cdot 0.836 \cdot 1.506$$

$$= 35.5 \text{ lbs. } [16.1 \text{ Kgf}]$$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
CH. COVER	1050.0	0.000	0.000	0.000	1.035	1.000	161.849
CH. SHELL	1050.0	0.000	0.000	0.000	1.035	1.000	161.849
CH. FLANGE	1050.0	0.000	0.000	0.000	1.035	1.000	161.849
SH. FLANGE	1050.0	0.000	0.000	0.000	1.035	1.000	161.849
MAIN SHELL	1050.0	0.000	0.000	0.000	1.035	1.000	161.849
SH. COVER	1050.0	0.000	0.000	0.000	1.035	1.000	161.849

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	1050	830.4	1398.93	161.849	17.6903
20	30	1050	830.4	4152	161.849	52.5046
30	40	1050	792	792	161.849	10.0153
40	50	1050	792	792	161.849	10.0153
50	60	1050	840	40320	161.849	509.871
60	70	1050	840	1437.09	161.849	18.1729

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-22	
Importance Factor		1.500	
Force Modification Factor R		3.000	
Site Class		D	
Component Elevation Ratio	z/h	0.000	
Amplification Factor	Ap	0.000	
Force Factor		0.000	
Consider Vertical Acceleration		Yes	
Minimum Acceleration Multiplier		0.000	
User Value of Sds (used if > 0)		0.000	
User Value of Sd1 (used if > 0)		0.000	
Component resonance duct. fact.	[CAR]	0.000	
Component strength factor	[Rpo]	0.000	
Overstrength factor	[Omega0]	0.000	
Response Acceleration Factor	[SMS]	0.326	
Response Acceleration Factor	[SM1]	0.139	
Building Importance Factor	[Ie]	1.000	
Site Latitude		0.000	deg
Site Longitude		0.000	deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.326 = 0.217$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.139 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(3.54330.75)$
 = 0.052 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33.0)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.052$, 0.03) per 12.8.2
 = 0.030

As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.217 \cdot 7026 \cdot 1.5$
 = 687.165 Kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.217 \cdot 7026$
 = 305.4 Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	330	878.279	85.8957
20	30	330	878.279	85.8957
30	40	330	878.279	85.8957
40	50	330	878.279	85.8957
50	Sad1	330	878.279	85.8957
Sad1	60	330	878.279	85.8957



50	60	330	878.279	85.8957
60	70	330	878.279	85.8957

The Earthquake Weight does not include the weight of the saddles.

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Shop/Field Installation Options :

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	3238.175 mm.
Center of Gravity of the Liquid	2710.691 mm.
Center of Gravity of the Nozzles	2338.657 mm.
Center of Gravity of the Tubesheet(s)	695.675 mm.
Center of Gravity of the Tubes	2953.175 mm.
Center of Gravity of the Baffles	3309.096 mm.
Center of Gravity of Bare Shell New and Cold	2773.511 mm.
Center of Gravity of Bare Shell Corroded	2782.412 mm.
Vessel CG in the Operating Condition	2730.563 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	2718.318 mm.
Vessel CG in the Test Condition	2729.320 mm.

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ASME Horizontal Vessel Analysis: Bundle pullout, Left Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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ASME Horizontal Vessel Analysis: Bundle pullout, Left Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Note:

Wear Pad Width (160.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

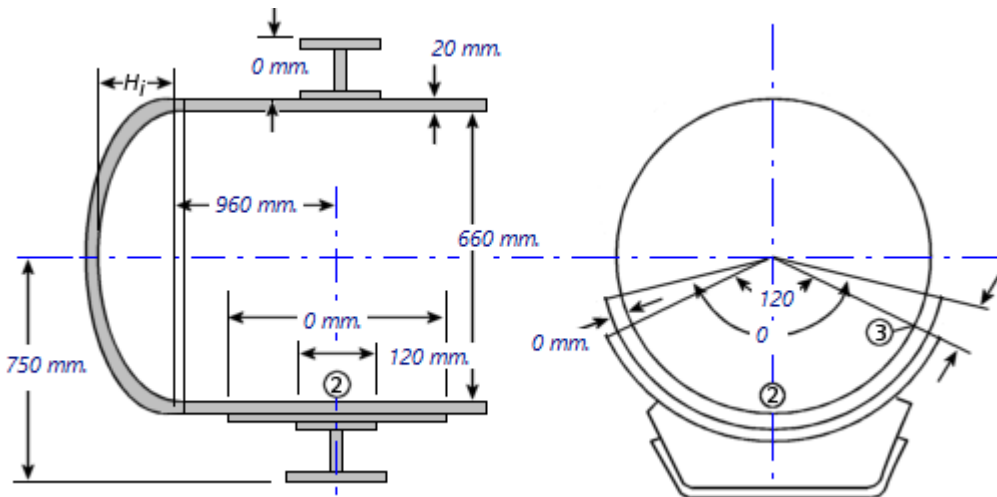
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(120.0 + 1.56 \cdot \sqrt{341.0 \cdot 18.0}, 2 \cdot 960.0)$$

$$= 242.2188 \text{ mm.}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	341.00	mm.
Shell Thickness used in this Case	t	20.000	mm.
Stiffened Vessel Length per 4.15.6	L	4950.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	960.00	mm.
Saddle Width	b1 or b	120.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	750.00	mm.
Coefficient of Friction	mu	0.60	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		6164.67	Kgf
Pressure used in Analysis	P	32.532	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	28.16	138.00
Long. Stress at Bottom of Midspan	33.48	138.00
Long. Stress at Top of Saddles	45.86	138.00
Long. Stress at Bottom of Saddles	22.48	138.00
Tangential Shear in Shell	7.06	110.40
Circ. Stress at Horn of Saddle	18.26	172.50
Circ. Compressive Stress in Shell	1.05	138.00



Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (618.3/2 + 0) \cdot 750.0/627.0024$
 $= 1109.3 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(98.66, 0) \cdot 750.0/2880.0$
 $= 25.7 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(687.17, 0) \cdot 750.0/2880.0$
 $= 178.9 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (687/2 + 0) \cdot 750.0/627.0024$
 $= 1232.9 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 4932 + \max(26, 1109, 179, 1233)$
 $= 6164.7 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587.169 (0.0 + \pi/4 (0.7 \cdot 1.2)^2)$
 $= 967.537 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	6242.69	Kgf
Transverse Shear Load	343.58	Kgf
Longitudinal Shear Load	687.17	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(660.0/2 + 20.0 + 12.0) \sin(120.0/2), 2 \cdot 341.0)$
 $= 627.002 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
 $= -6165 \cdot 960.0 [1 - (1 - 960.0/4950.0 + (341.0^2 - 0.0^2)/(2 \cdot 960.0 \cdot 4950.0))/(1 + (4 \cdot 0.0)/(3 \cdot 4950.0))]$
 $= -1075.4 \text{ Kg-m}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 6165 \cdot 4950/4 ((1 + 2(341^2 - 0^2)/(4950^2))/(1 + (4 \cdot 0)/(3 \cdot 4950)) - 4 \cdot 960/4950)$
 $= 1783.1 \text{ Kg-m}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$
 $= 32.532 \cdot 341.0/(2 \cdot 18.0) - 1783.1/(\pi \cdot 341.0^2 \cdot 18.0)$
 $= 28.16 \text{ N/mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:



$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 32.532 \cdot 341.0 / (2 \cdot 18.0) + 1783.1 / (\pi \cdot 341.0^2 \cdot 18.0)$$

$$= 33.48 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 32.532 \cdot 341.0 / (2 \cdot 18.0) - 1075.4 / (0.1066 \cdot \pi \cdot 341.0^2 \cdot 18.0)$$

$$= 45.86 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 32.532 \cdot 341.0 / (2 \cdot 18.0) + 1075.4 / (0.1923 \cdot \pi \cdot 341.0^2 \cdot 18.0)$$

$$= 22.48 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot hm / 3))$$

$$= 6165(4950.0 - 2 \cdot 960.0) / (4950.0 + (4 \cdot 0.0 / 3))$$

$$= 3773.5 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [$\tau2$]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 3773.52 / (341.0 \cdot 18.0)$$

$$= 7.06 \text{ N./mm}^2$$

Decay Length (4.15.20) [$x1, x2$]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(341.0 \cdot 18.0)}$$

$$= 61.109 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [$\sigma6$]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 6165 \cdot 0.1 / (18.0(120.0 + 61.11 + 61.11))$$

$$= -1.05 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [$\sigma7$]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -6165 / (4 \cdot 18.0(120.0 + 61.109 + 61.109)) - 3 \cdot 0.0529 \cdot 6165 / (2 \cdot 18.0^2)$$

$$= -18.26 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(120.0 + 1.56 \cdot \sqrt{(341.0 \cdot 18.0)}, 2 \cdot 960.0)$$

$$= 242.22 \text{ mm.}$$

Distance between Saddle Supports [Ls]:

$$= 2880.0 \text{ mm.}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot Ls(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000012 \cdot 2880.0(145.0 - 21.1)$$

$$= 4.422 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	690.0000	mm.
Baseplate Thickness	Bpthk	16.0000	mm.
Baseplate Width	Bpwid	140.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	12.0000	mm.
Web Thickness	Webtk	12.0000	mm.
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	232.6	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.600	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(332.0 \cdot 18.0)} + 160.0) 18.0^3) / 12$$

$$= 13.637 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 13.637 + 50.507 (144.366 - 9.0)^2$$

$$= 9268.496 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	280.6	18.0	9.0	50.5	45456.4	0.927E+04
Wearplate	160.0	12.0	24.0	19.2	46080.0	0.278E+04
Web	12.0	372.0	216.0	44.6	964223.9	0.744E+04
BasePlate	140.0	16.0	410.0	22.4	918399.9	0.158E+05
Totals	...	...	...	136.7	1974160.2	0.353E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1974160 / 136.747$$

$$= 144.366 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 6164.668$$

$$= 1254.6437 \text{ Kgf}$$

Tension Stress, St = (Fh/As) = 1.4267 N./mm²

Allowed Stress, Sa = 0.6 * Yield Str = 139.5584 N./mm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 750.0 - 332.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 475.438 \text{ mm.}$$

Bending Moment, M = Fh * d = 596.5173 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 2.3923 N./mm²

Allowed Stress, Sa = 2/3 * Yield Str = 155.0649 N./mm²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(6165 + 78)140.0 / (2 \cdot 690.0 \cdot 155.065)}$$

$$= 10.962 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (330.0 + 20.0 + 12.0)$$

$$= 627.002 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 627.0024 / (3 - 1)$$

$$= 313.501 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 313.5012 \cdot 140.0/2$$



$$= 219.451 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 6242.686 / (690.0 \cdot 140.0)$$

$$= 6.462 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 219.5 \cdot 6.46$$

$$= 1418.181 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 12.96 + 18.81$$

$$= 31.770 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 1418.2 / 31.7701$$

$$= 4.378 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	12.0	108.0	66.0	13.0	85536.0	290.
Web	156.8	12.0	6.0	18.8	11286.0	115.
Totals	...	...	...	31.8	96822.0	404.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 120.0 - 12.0$$

$$= 108.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 96822.039 / 31.77$$

$$= 30.476 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(30.476, 120.0 - 30.476)$$

$$= 89.524 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(404.5 / 31.77)}$$

$$= 35.680 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 750.0 - \cos(120.0/2) (330.0 + 20.0 + 12.0) - 16.0$$

$$= 553.000 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19536\text{E}+09 / 232.6)}$$

$$= 128.763$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 553.0 / 35.68$$

$$= 15.499$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 687.2 / (2 \cdot 690.0) \cdot 313.501 \cdot 553.0 / 2$$

= 43.164 Kg-m.

Compressive Allowable, $KL/r < Cc$ (15.4987 < 128.7632) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (15.5)^2 / (2 \cdot 128.76^2)) 233 / (5/3 + 3 \cdot (15.5) / (8 \cdot 128.76) - (15.5^3) / (8 \cdot 128.76^3))$$

$$= 134.9 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 4.38/134.91 + (43.16 \cdot 89.524/4044641) / 155.06$$

$$= 0.093$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	12.0	108.0	66.0	13.0	85536.0	384.
Web	313.5	12.0	6.0	37.6	22572.1	93.4
Totals	...	...	...	50.6	108108.1	477.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 108108.086/50.58$$

$$= 21.374 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(21.374, \text{abs}(120.0 - 21.374))$$

$$= 98.626 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 750.0 - \cos(362.0/2) (16.0 + 0.0 + 0.0) - 0.0$$

$$= 372.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (477.5/50.58) }$$

$$= 30.725 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 372.0/30.725$$

$$= 12.107$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 687.165 / (2 \cdot 690.0)$$

$$= 0.498 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.498 \cdot 313.501 \cdot 372.0$$

$$= 58.073 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 58.073/48415.07$$

$$= 11.763 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < Cc$ (12.1073 < 128.7632) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (12.11)^2 / (2 \cdot 128.76^2)) 233 / (5/3 + 3 \cdot (12.11) / (8 \cdot 128.76) - (12.11^3) / (8 \cdot 128.76^3))$$

$$= 136.1 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$



$$= 5.57/136.07 + (58.07 \cdot 98.626/477.5)/155.06$$

$$= 0.117$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	150.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	70.0 mm.
Nominal Bolt Diameter	Bnd	30.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BlArea	3.9088 cm ²
Saddle Load QO (Weight)	QO	5009.7 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	178.9 Kg
Maximum Transverse Force	Ft	343.6 Kg
Maximum Longitudinal Force	Fl	687.2 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 687.2 / (3.909 \cdot 2)$$

$$= 8.6 \text{ N./mm}^2. \text{ Must be less than } 100.0 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 343.6 / (3.909 \cdot 2)$$

$$= 4.3 \text{ N./mm}^2. \text{ Must be less than } 100.0 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 687.17 / (100.0 \cdot 2.0)$$

$$= 0.3369 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 750.0 \cdot 343.58 + 0.0$$

$$= 257.69 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 257.69/5009.74$$

$$= 51.44 \text{ mm.} < B_{plen}/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$$

$$= \max[0.0, 0.3369, 0.0]$$

$$= 0.3369 \text{ cm}^2$$

MDMT Calculations for: FIXED SADDLE, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 12.0, c = 2.0 mm., E* = 1.0

Thickness Ratio = tr * E*/(tg - c) = 0.46, Temp. Reduction = 38 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B

-23 °C



Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (160.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(120.0 + 1.56 \cdot \sqrt{341.0 \cdot 18.0}, 2 \cdot 960.0)$$

$$= 242.2188 \text{ mm.}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	341.00	mm.
Shell Thickness used in this Case	t	20.000	mm.
Stiffened Vessel Length per 4.15.6	L	4950.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	960.00	mm.
Saddle Width	b1 or b	120.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	176.00	mm.
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	750.00	mm.
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		3632.86	Kgf
Pressure used in Analysis	P	32.532	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	29.57	138.00
Long. Stress at Bottom of Midspan	32.06	138.00
Long. Stress at Top of Saddles	41.64	138.00
Long. Stress at Bottom of Saddles	24.82	138.00
Tangential Shear in Shell	3.97	110.40
Circ. Stress at Horn of Saddle	10.76	172.50
Circ. Compressive Stress in Shell	0.62	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0 (618.3 / 2 + 0) \cdot 750.0 / 627.0024$$

$$= 1109.3 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(98.66, 0) \cdot 750.0 / 2880.0$$

$$= 25.7 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(687.17, 0) \cdot 750.0 / 2880.0$$

$$= 178.9 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(687/2 + 0) \cdot 750.0/627.0024$$

$$= 1232.9 \text{ Kg}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 2400 + \max(26, 1109, 179, 1233)$$

$$= 3632.9 \text{ Kg}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587.169(0.0 + \pi/4(0.7 \cdot 1.2)^2)$$

$$= 967.537 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	3710.88	Kgf
Transverse Shear Load	343.58	Kgf
Longitudinal Shear Load	687.17	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(660.0/2 + 20.0 + 12.0)\sin(120.0/2), 2 \cdot 341.0)$$

$$= 627.002 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -3633 \cdot 960.0 [1 - (1 - 960.0/4950.0 + (341.0^2 - 176.0^2)/(2 \cdot 960.0 \cdot 4950.0))/(1 + (4 \cdot 176.0)/(3 \cdot 4950.0))]$$

$$= -773.7 \text{ Kg-m}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 3633 \cdot 4950/4((1 + 2(341^2 - 176^2)/(4950^2))/(1 + (4 \cdot 176)/(3 \cdot 4950)) - 4 \cdot 960/4950)$$

$$= 834.5 \text{ Kg-m}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 32.532 \cdot 341.0/(2 \cdot 18.0) - 834.5/(\pi \cdot 341.0^2 \cdot 18.0)$$

$$= 29.57 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 32.532 \cdot 341.0/(2 \cdot 18.0) + 834.5/(\pi \cdot 341.0^2 \cdot 18.0)$$

$$= 32.06 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 32.532 \cdot 341.0/(2 \cdot 18.0) - 773.7/(0.1066 \cdot \pi \cdot 341.0^2 \cdot 18.0)$$

$$= 41.64 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m/(2t) + M1/(K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 32.532 \cdot 341.0/(2 \cdot 18.0) + 773.7/(0.1923 \cdot \pi \cdot 341.0^2 \cdot 18.0)$$

$$= 24.82 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 3633(4950.0 - 2 \cdot 960.0)/(4950.0 + (4 \cdot 176.0/3))$$

$$= 2123.1 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 2123.1 / (341.0 \cdot 18.0)$$

$$= 3.97 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(341.0 \cdot 18.0)}$$

$$= 61.109 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 3633 \cdot 0.1 / (18.0(120.0 + 61.11 + 61.11))$$

$$= -0.62 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -3633 / (4 \cdot 18.0(120.0 + 61.109 + 61.109)) -$$

$$3 \cdot 0.0529 \cdot 3633 / (2 \cdot 18.0^2)$$

$$= -10.76 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(120.0 + 1.56 \cdot \sqrt{(341.0 \cdot 18.0)}, 2 \cdot 960.0)$$

$$= 242.22 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	690.0000	mm.
Baseplate Thickness	Bpthk	16.0000	mm.
Baseplate Width	Bpwid	140.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	12.0000	mm.
Web Thickness	Webtk	12.0000	mm.
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	232.6	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(332.0 \cdot 18.0)} + 160.0)18.0^3) / 12$$

$$= 13.637 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 13.637 + 50.507(144.366 - 9.0)^2$$

$$= 9268.496 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	280.6	18.0	9.0	50.5	45456.4	0.927E+04
Wearplate	160.0	12.0	24.0	19.2	46080.0	0.278E+04
Web	12.0	372.0	216.0	44.6	964223.9	0.744E+04
BasePlate	140.0	16.0	410.0	22.4	918399.9	0.158E+05
Totals	...	...	...	136.7	1974160.2	0.353E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1974160 / 136.747$$

$$= 144.366 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$



Saddle Splitting Coefficient [K1]:

$$\begin{aligned} &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\ &= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0)) \\ &= 0.2035 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned} &= K1 \cdot Q \\ &= 0.204 \cdot 3632.863 \\ &= 739.3665 \text{ Kgf} \end{aligned}$$

$$\begin{aligned} \text{Tension Stress, } St &= (Fh / As) = 0.8408 \text{ N./mm}^2 \\ \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 139.5584 \text{ N./mm}^2 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned} &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\ &= 750.0 - 332.0 \cdot \sin(120.0/2) / 1.0472 \\ &= 475.438 \text{ mm.} \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 351.5300 \text{ Kg-m.}$$

$$\begin{aligned} \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 1.4098 \text{ N./mm}^2 \\ \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 155.0649 \text{ N./mm}^2 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned} &= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\ &= \sqrt{(3(3633 + 78)140.0) / (2 \cdot 690.0 \cdot 155.065)} \\ &= 8.451 \text{ mm.} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned} &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 \cdot \cos(90 - 120.0/2) (330.0 + 20.0 + 12.0) \\ &= 627.002 \text{ mm.} \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned} &= \text{Web Length} / (\text{Nr ribs} - 1) \\ &= 627.0024 / (3 - 1) \\ &= 313.501 \text{ mm.} \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot \text{Bpwid} / 2 \\ &= 313.5012 \cdot 140.0 / 2 \\ &= 219.451 \text{ cm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\ &= 3710.881 / (690.0 \cdot 140.0) \\ &= 3.841 \text{ Kg/cm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= Ap \cdot Bp \\ &= 219.5 \cdot 3.84 \\ &= 843.018 \text{ Kgf} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= \text{Rib Area} + \text{Web Area} \\ &= 12.96 + 18.81 \\ &= 31.770 \text{ cm}^2 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned} &= P / Ar \\ &= 843.0 / 31.7701 \\ &= 2.602 \text{ N./mm}^2 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	12.0	108.0	66.0	13.0	85536.0	290.
Web	156.8	12.0	6.0	18.8	11286.0	115.
Totals	...	...	...	31.8	96822.0	404.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 120.0 - 12.0$$

$$= 108.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 96822.039/31.77$$

$$= 30.476 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(30.476, 120.0 - 30.476)$$

$$= 89.524 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{404.5/31.77}$$

$$= 35.680 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 750.0 - \cos(120.0/2) (330.0 + 20.0 + 12.0) - 16.0$$

$$= 553.000 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19536\text{E}+09/232.6)}$$

$$= 128.763$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 553.0/35.68$$

$$= 15.499$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 687.2 / (2 \cdot 690.0) \cdot 313.501 \cdot 553.0/2$$

$$= 43.164 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($15.4987 < 128.7632$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (15.5)^2 / (2 \cdot 128.76^2)) 233 /$$

$$(5/3 + 3 \cdot (15.5) / (8 \cdot 128.76) - (15.5)^3 / (8 \cdot 128.76^3))$$

$$= 134.9 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 2.6/134.91 + (43.16 \cdot 89.524/4044641) / 155.06$$

$$= 0.080$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	12.0	108.0	66.0	13.0	85536.0	384.
Web	313.5	12.0	6.0	37.6	22572.1	93.4
Totals	...	...	...	50.6	108108.1	477.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 108108.086/50.58$$

$$= 21.374 \text{ mm.}$$



Distance to Centroid [C1]:
 $= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$
 $= \max(21.374, \text{abs}(120.0 - 21.374))$
 $= 98.626 \text{ mm.}$

Length of Inner Rib [L]:
 $= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$
 $= 750.0 - \cos(362.0/2)(16.0 + 0.0 + 0.0) - 0.0$
 $= 372.000 \text{ mm.}$

Radius of Gyration [r]:
 $= \sqrt{ \text{Total Inertia} / \text{Total Area} }$
 $= \sqrt{ 477.5/50.58 }$
 $= 30.725 \text{ mm.}$

Slenderness ratio [KL/r]:
 $= KL/r$
 $= 1 \cdot 372.0/30.725$
 $= 12.107$

Unit Force [Force,u]:
 $= FI / (2 \cdot \text{Baseplate Length})$
 $= 687.165/(2 \cdot 690.0)$
 $= 0.498 \text{ Kg/mm.}$

Moment at base of inner Rib [Mbase,c]:
 $= \text{Unit Force} \cdot e \cdot L$
 $= 0.498 \cdot 313.501 \cdot 372.0$
 $= 58.073 \text{ Kg-m.}$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 $= \text{Bending Moment} / \text{Section Modulus}$
 $= 58.073/48415.07$
 $= 11.763 \text{ N./mm}^2$

Compressive Allowable, $KL/r < C_c$ ($12.1073 < 128.7632$) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
 $= (1 - (12.11)^2 / (2 \cdot 128.76^2)) 233 /$
 $(5/3 + 3 \cdot (12.11) / (8 \cdot 128.76) - (12.11^3) / (8 \cdot 128.76^3))$
 $= 136.1 \text{ N./mm}^2$

AISC Unity Check of Inside Ribs (must be ≤ 1)
 $= S_c/S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$
 $= 3.31/136.07 + (58.07 \cdot 98.626/477.5) / 155.06$
 $= 0.100$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	150.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	70.0 mm.
Nominal Bolt Diameter	Bnd	30.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BlArea	3.9088 cm ²
Saddle Load Q0 (Weight)	Q0	2477.9 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	178.9 Kg
Maximum Transverse Force	Ft	343.6 Kg
Maximum Longitudinal Force	Fl	687.2 Kg
Saddle Bolted to Steel Foundation	No	No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:
 $= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 687.2 / (3.909 \cdot 2)$
 $= 8.6 \text{ N./mm}^2$. Must be less than 100.0 N./mm².



Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $343.6 / (3.909 \cdot 2)$
= 4.3 N./mm^2 . Must be less than 100.0 N./mm^2 .

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
= $687.17 / (100.0 \cdot 2.0)$
= 0.3369 cm^2

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $750.0 \cdot 343.58 + 0.0$
= 257.69 Kg-m .

Eccentricity (e):
= R_{mom} / QO
= $257.69 / 2477.93$
= 103.99 mm . < $B_{plen}/6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[Bltarearl, Bltarears, Bltareart]$
= $\max[0.0, 0.3369, 0.0]$
= 0.3369 cm^2

MDMT Calculations for: RHS Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 12.0, c = 2.0 mm., $E^* = 1.0$
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.46$, Temp. Reduction = 38°C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (160.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

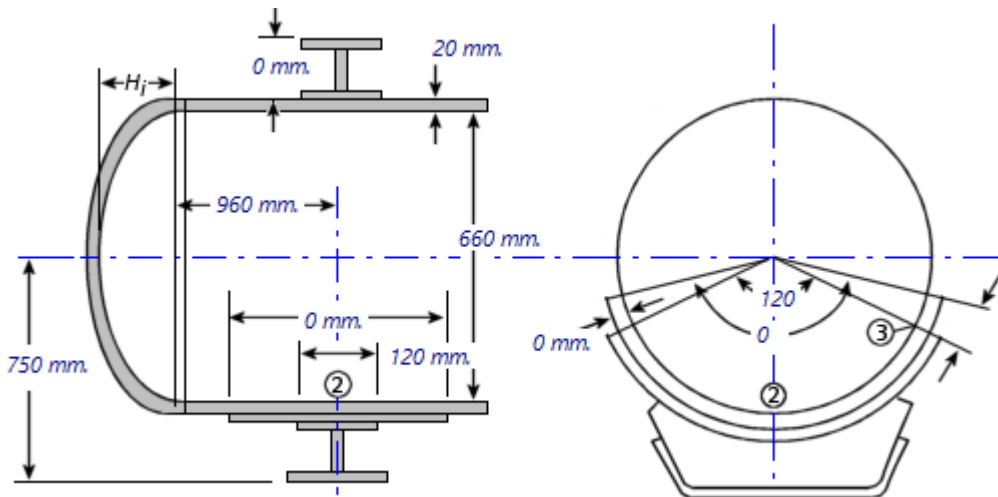
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(120.0 + 1.56 \cdot \sqrt{341.0 \cdot 18.0}, 2 \cdot 960.0)$$

$$= 242.2188 \text{ mm.}$$

Input and Calculated Values:

Vessel Mean Radius	R_m	341.00	mm.
Shell Thickness used in this Case	t	20.000	mm.
Stiffened Vessel Length per 4.15.6	L	4950.00	mm.
Distance from Saddle to Vessel tangent	a_1 or a	960.00	mm.
Saddle Width	b_1 or b	120.00	mm.
Saddle Bearing Angle	δ or θ	120.00	degrees
Shell Allowable Stress used in Calculation		247.00	N./mm ²
Head Allowable Stress used in Calculation		247.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	750.00	mm.
Coefficient of Friction	μ	0.60	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		5306.46	Kgf
Pressure used in Analysis	P	44.079	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	39.47	247.00
Long. Stress at Bottom of Midspan	44.04	247.00
Long. Stress at Top of Saddles	54.70	247.00
Long. Stress at Bottom of Saddles	34.58	247.00
Tangential Shear in Shell	6.08	138.58
Circ. Stress at Horn of Saddle	15.72	370.49
Circ. Compressive Stress in Shell	0.91	247.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (204.0/2 + 0) \cdot 750.0/627.0024$
 $= 366.1 \text{ Kg}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(32.56, 0) \cdot 750.0/2880.0$
 $= 8.5 \text{ Kg}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 4940 + \max(8, 366, 0, 0)$
 $= 5306.5 \text{ Kg}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587.169 (0.0 + \pi/4 (0.7 \cdot 1.2)^2)$
 $= 967.537 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	5384.47	Kg
Transverse Shear Load	102.01	Kg
Longitudinal Shear Load	32.56	Kg

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(660.0/2 + 20.0 + 12.0) \sin(120.0/2), 2 \cdot 341.0)$
 $= 627.002 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$
 $= -5306 \cdot 960.0 [1 - (1 - 960.0/4950.0 + (341.0^2 - 0.0^2) / (2 \cdot 960.0 \cdot 4950.0)) / (1 + (4 \cdot 0.0) / (3 \cdot 4950.0))]$
 $= -925.7 \text{ Kg-m}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L / 4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 5306 \cdot 4950 / 4 ((1 + 2(341^2 - 0^2)/(4950^2)) / (1 + (4 \cdot 0)/(3 \cdot 4950)) - 4 \cdot 960/4950)$
 $= 1534.9 \text{ Kg-m}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$
 $= 44.079 \cdot 341.0 / (2 \cdot 18.0) - 1534.9 / (\pi \cdot 341.0^2 \cdot 18.0)$
 $= 39.47 \text{ N/mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$
 $= 44.079 \cdot 341.0 / (2 \cdot 18.0) + 1534.9 / (\pi \cdot 341.0^2 \cdot 18.0)$
 $= 44.04 \text{ N/mm}^2$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:
 $= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$
 $= 44.079 \cdot 341.0 / (2 \cdot 18.0) - 925.7 / (0.1066 \cdot \pi \cdot 341.0^2 \cdot 18.0)$
 $= 54.70 \text{ N/mm}^2$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 44.079 \cdot 341.0 / (2 \cdot 18.0) + 925.7 / (0.1923 \cdot \pi \cdot 341.0^2 \cdot 18.0)$$

$$= 34.58 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 5306(4950.0 - 2 \cdot 960.0) / (4950.0 + (4 \cdot 0.0 / 3))$$

$$= 3248.2 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 3248.19 / (341.0 \cdot 18.0)$$

$$= 6.08 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(341.0 \cdot 18.0)}$$

$$= 61.109 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 5306 \cdot 0.1 / (18.0(120.0 + 61.11 + 61.11))$$

$$= -0.91 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -5306 / (4 \cdot 18.0(120.0 + 61.109 + 61.109)) - 3 \cdot 0.0529 \cdot 5306 / (2 \cdot 18.0^2)$$

$$= -15.72 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(120.0 + 1.56 \cdot \sqrt{(341.0 \cdot 18.0)}, 2 \cdot 960.0)$$

$$= 242.22 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	690.0000	mm.
Baseplate Thickness	Bpthk	16.0000	mm.
Baseplate Width	Bpwid	140.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	12.0000	mm.
Web Thickness	Webtk	12.0000	mm.
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.600	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(332.0 \cdot 18.0)} + 160.0)^3) / 12$$

$$= 13.637 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 13.637 + 50.507(144.366 - 9.0)^2$$

$$= 9268.496 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ² cm ⁴
Shell	280.6	18.0	9.0	50.5	45456.4	0.927E+04
Wearplate	160.0	12.0	24.0	19.2	46080.0	0.278E+04
Web	12.0	372.0	216.0	44.6	964223.9	0.744E+04
BasePlate	140.0	16.0	410.0	22.4	918399.9	0.158E+05
Totals	...	...	...	136.7	1974160.2	0.353E+05

Distance to Centroid [C1]:

$$= AY / A$$



$$= 1974160/136.747$$
$$= 144.366 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$
$$= 180 - 120.0/2$$
$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$
$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$
$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$
$$= 0.204 \cdot 5306.456$$
$$= 1079.9789 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh/As) = 1.2281 \text{ N./mm}^2$$
$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$
$$= 750.0 - 332.0 \cdot \sin(120.0/2) / 1.0472$$
$$= 475.438 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 513.4734 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 2.0592 \text{ N./mm}^2$$
$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$
$$= \sqrt[3]{(3(5306 + 78)140.0) / (2 \cdot 690.0 \cdot 173.33)}$$
$$= 9.629 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$
$$= 2 \cdot \cos(90 - 120.0/2) (330.0 + 20.0 + 12.0)$$
$$= 627.002 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$
$$= 627.0024 / (3 - 1)$$
$$= 313.501 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$
$$= 313.5012 \cdot 140.0/2$$
$$= 219.451 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$
$$= 5384.474 / (690.0 \cdot 140.0)$$
$$= 5.574 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$
$$= 219.5 \cdot 5.57$$
$$= 1223.217 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$
$$= 12.96 + 18.81$$
$$= 31.770 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

= 1223.2/31.7701
 = 3.776 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	12.0	108.0	66.0	13.0	85536.0	290.
Web	156.8	12.0	6.0	18.8	11286.0	115.
Totals	...	...	...	31.8	96822.0	404.

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 120.0 - 12.0
 = 108.000 mm.

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 96822.039/31.77
 = 30.476 mm.

Distance to Centroid [C1]:
 = max(ytot, Saddle Width - ytot)
 = max(30.476, 120.0 - 30.476)
 = 89.524 mm.

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{404.5/31.77}$
 = 35.680 mm.

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 750.0 - cos(120.0/2)(330.0 + 20.0 + 12.0) - 16.0
 = 553.000 mm.

Intermediate Term [Cc]:
 $= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 $= \sqrt{2 \cdot \pi^2 \cdot 0.19536\text{E}+09/260.0}$
 = 121.790

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 553.0/35.68
 = 15.499

Bending Moment [Rm]:
 $= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$
 $= 32.6 / (2 \cdot 690.0) \cdot 313.501 \cdot 553.0/2$
 = 2.045 Kg-m.

Compressive Allowable, KL/r < Cc (15.4987 < 121.79) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 $= (1 - (15.5)^2 / (2 \cdot 121.79^2)) 260 / (5/3 + 3 \cdot (15.5) / (8 \cdot 121.79) - (15.5^3) / (8 \cdot 121.79^3))$
 = 150.4 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
 $= Sc/Sca + (Rm \cdot C1 / I) / Sba$
 $= 3.78/150.45 + (2.05 \cdot 89.524/4044641) / 173.33$
 = 0.028

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	12.0	108.0	66.0	13.0	85536.0	384.



Web	313.5	12.0	6.0	37.6	22572.1	93.4
Totals	...	...	...	50.6	108108.1	477.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 108108.086/50.58$$

$$= 21.374 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(21.374, \text{abs}(120.0 - 21.374))$$

$$= 98.626 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 750.0 - \cos(362.0/2)(16.0 + 0.0 + 0.0) - 0.0$$

$$= 372.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(477.5/50.58)}$$

$$= 30.725 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 372.0/30.725$$

$$= 12.107$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 32.559/(2 \cdot 690.0)$$

$$= 0.024 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.024 \cdot 313.501 \cdot 372.0$$

$$= 2.752 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 2.752/48415.07$$

$$= 0.557 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (12.1073 < 121.79) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.11)^2 / (2 \cdot 121.79^2)) 260 /$$

$$(5/3 + 3 \cdot (12.11) / (8 \cdot 121.79) - (12.11^3) / (8 \cdot 121.79^3))$$

$$= 151.8 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 4.8/151.84 + (2.75 \cdot 98.626/477.5) / 173.33$$

$$= 0.035$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	70.0 mm.
Nominal Bolt Diameter	Bnd	30.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BlArea	3.9088 cm ²
Saddle Load Q0 (Weight)	Q0	5018.4 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	8.5 Kg
Maximum Transverse Force	Ft	102.0 Kg
Maximum Longitudinal Force	Fl	32.6 Kg

Saddle Bolted to Steel Foundation

No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
= $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $32.6 / (3.909 \cdot 2)$
= 0.4 N./mm^2 . Must be less than 103.2 N./mm^2 .

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $102.0 / (3.909 \cdot 2)$
= 1.3 N./mm^2 . Must be less than 103.2 N./mm^2 .

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbolls})$
= $32.56 / (103.2 \cdot 2.0)$
= 0.0155 cm^2

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $750.0 \cdot 102.01 + 0.0$
= 76.51 Kg-m .

Eccentricity (e):
= R_{mom} / QO
= $76.51 / 5018.39$
= 15.25 mm . < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
= $\max[0.0, 0.0155, 0.0]$
= 0.0155 cm^2

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
Wear Pad Width (160.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
= $\min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
= $\min(120.0 + 1.56 \cdot \sqrt{341.0 \cdot 18.0}, 2 \cdot 960.0)$
= 242.2188 mm .

Input and Calculated Values:

Vessel Mean Radius	Rm	341.00	mm.
Shell Thickness used in this Case	t	20.000	mm.
Stiffened Vessel Length per 4.15.6	L	4950.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	960.00	mm.
Saddle Width	b1 or b	120.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	176.00	mm.
Shell Allowable Stress used in Calculation		247.00	N./mm ²
Head Allowable Stress used in Calculation		247.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	



Distance from Saddle Base to Centerline B 750.00 mm.
 Coefficient of Friction mu 0.30

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces 2685.80 Kgf
 Pressure used in Analysis P 44.079 bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	40.84	247.00
Long. Stress at Bottom of Midspan	42.68	247.00
Long. Stress at Top of Saddles	49.76	247.00
Long. Stress at Bottom of Saddles	37.32	247.00
Tangential Shear in Shell	2.94	138.58
Circ. Stress at Horn of Saddle	7.96	370.49
Circ. Compressive Stress in Shell	0.46	247.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(204.0/2 + 0) \cdot 750.0/627.0024$
 $= 366.1 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,\text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(32.56, 0) \cdot 750.0/2880.0$
 $= 8.5 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 2320 + \max(8, 366, 0, 0)$
 $= 2685.8 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587.169(0.0 + \pi/4(0.7 \cdot 1.2)^2)$
 $= 967.537 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 2763.82 Kgf
 Transverse Shear Load 102.01 Kgf
 Longitudinal Shear Load 32.56 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(660.0/2 + 20.0 + 12.0)\sin(120.0/2), 2 \cdot 341.0)$
 $= 627.002 \text{ mm.}$

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011
 K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405
 K9 = 0.2711 K10 = 0.0581 K1* = 0.1923

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
 $= -2686 \cdot 960.0 [1 - (1 - 960.0/4950.0 + (341.0^2 - 176.0^2)/(2 \cdot 960.0 \cdot 4950.0))/(1 + (4 \cdot 176.0)/(3 \cdot 4950.0))]$
 $= -572.0 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a / L) \\
 &= 2686 \cdot 4950 / 4 \cdot ((1 + 2(341^2 - 176^2) / (4950^2)) / (1 + (4 \cdot 176) / (3 \cdot 4950)) - 4 \cdot 960 / 4950) \\
 &= 617.0 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= 44.079 \cdot 341.0 / (2 \cdot 18.0) - 617.0 / (\pi \cdot 341.0^2 \cdot 18.0) \\
 &= 40.84 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= 44.079 \cdot 341.0 / (2 \cdot 18.0) + 617.0 / (\pi \cdot 341.0^2 \cdot 18.0) \\
 &= 42.68 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 44.079 \cdot 341.0 / (2 \cdot 18.0) - 572.0 / (0.1066 \cdot \pi \cdot 341.0^2 \cdot 18.0) \\
 &= 49.76 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 44.079 \cdot 341.0 / (2 \cdot 18.0) + 572.0 / (0.1923 \cdot \pi \cdot 341.0^2 \cdot 18.0) \\
 &= 37.32 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a) / (L + (4 \cdot hm / 3)) \\
 &= 2686(4950.0 - 2 \cdot 960.0) / (4950.0 + (4 \cdot 176.0 / 3)) \\
 &= 1569.6 \text{ Kgf}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (Rm \cdot t) \\
 &= 1.1707 \cdot 1569.63 / (341.0 \cdot 18.0) \\
 &= 2.94 \text{ N./mm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(Rm \cdot t)} \\
 &= 0.78 \cdot \sqrt{(341.0 \cdot 18.0)} \\
 &= 61.109 \text{ mm.}
 \end{aligned}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (t(b + X1 + X2)) \\
 &= -0.7603 \cdot 2686 \cdot 0.1 / (18.0(120.0 + 61.11 + 61.11)) \\
 &= -0.46 \text{ N./mm}^2
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [σ_7]:

$$\begin{aligned}
 &= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2) \\
 &= -2686 / (4 \cdot 18.0(120.0 + 61.109 + 61.109)) - \\
 &\quad 3 \cdot 0.0529 \cdot 2686 / (2 \cdot 18.0^2) \\
 &= -7.96 \text{ N./mm}^2
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\
 &= \min(120.0 + 1.56 \cdot \sqrt{(341.0 \cdot 18.0)}, 2 \cdot 960.0) \\
 &= 242.22 \text{ mm.}
 \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	690.0000	mm.
Baseplate Thickness	Bpthk	16.0000	mm.
Baseplate Width	Bpwid	140.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	12.0000	mm.
Web Thickness	Webtk	12.0000	mm.
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(332.0 \cdot 18.0)} + 160.0)^3) / 12$$

$$= 13.637 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 13.637 + 50.507 (144.366 - 9.0)^2$$

$$= 9268.496 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	280.6	18.0	9.0	50.5	45456.4	0.927E+04
Wearplate	160.0	12.0	24.0	19.2	46080.0	0.278E+04
Web	12.0	372.0	216.0	44.6	964223.9	0.744E+04
BasePlate	140.0	16.0	410.0	22.4	918399.9	0.158E+05
Totals	...	...	...	136.7	1974160.2	0.353E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1974160 / 136.747$$

$$= 144.366 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 2685.803$$

$$= 546.6192 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.6216 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 750.0 - 332.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 475.438 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 259.8888 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 1.0423 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(2686 + 78)140.0) / (2 \cdot 690.0 \cdot 173.33)}$$

$$= 6.899 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (330.0 + 20.0 + 12.0)$$

$$= 627.002 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 627.0024 / (3 - 1)$$

$$= 313.501 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 313.5012 \cdot 140.0 / 2$$

$$= 219.451 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 2763.821 / (690.0 \cdot 140.0)$$

$$= 2.861 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 219.5 \cdot 2.86$$

$$= 627.870 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 12.96 + 18.81$$

$$= 31.770 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 627.9 / 31.7701$$

$$= 1.938 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	12.0	108.0	66.0	13.0	85536.0	290.
Web	156.8	12.0	6.0	18.8	11286.0	115.
Totals	...	...	...	31.8	96822.0	404.

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 120.0 - 12.0$$

$$= 108.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 96822.039 / 31.77$$

$$= 30.476 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(ytot, Saddle \text{ Width} - ytot)$$

$$= \max(30.476, 120.0 - 30.476)$$

$$= 89.524 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (404.5 / 31.77) }$$

$$= 35.680 \text{ mm.}$$

Length of Outer Rib [L]:

$$= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 750.0 - \cos(120.0 / 2) (330.0 + 20.0 + 12.0) - 16.0$$

$$= 553.000 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 0.19536E+09 / 260.0) }$$

$$= 121.790$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 553.0 / 35.68$$

$$= 15.499$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 32.6 / (2 \cdot 690.0) \cdot 313.501 \cdot 553.0 / 2$$

$$= 2.045 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ (15.4987 < 121.79) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (15.5)^2 / (2 \cdot 121.79^2)) 260 /$$

$$(5/3 + 3 \cdot (15.5) / (8 \cdot 121.79) - (15.5^3) / (8 \cdot 121.79^3))$$

$$= 150.4 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 1.94 / 150.45 + (2.05 \cdot 89.524 / 4044641) / 173.33$$

$$= 0.015$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	12.0	108.0	66.0	13.0	85536.0	384.
Web	313.5	12.0	6.0	37.6	22572.1	93.4
Totals	...	...	...	50.6	108108.1	477.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 108108.086 / 50.58$$

$$= 21.374 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(21.374, \text{abs}(120.0 - 21.374))$$

$$= 98.626 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 750.0 - \cos(362.0 / 2) (16.0 + 0.0 + 0.0) - 0.0$$

$$= 372.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (477.5 / 50.58) }$$

$$= 30.725 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 372.0 / 30.725$$

$$= 12.107$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 32.559 / (2 \cdot 690.0)$$

$$= 0.024 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.024 \cdot 313.501 \cdot 372.0$$

$$= 2.752 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 2.752 / 48415.07$$

$$= 0.557 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (12.1073 < 121.79) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.11)^2 / (2 \cdot 121.79^2)) 260 /$$

$$(5/3 + 3 \cdot (12.11) / (8 \cdot 121.79) - (12.11^3) / (8 \cdot 121.79^3))$$

$$= 151.8 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I)/Sba$$

$$= 2.47/151.84 + (2.75 \cdot 98.626/477.5)/173.33$$

$$= 0.019$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	70.0 mm.
Nominal Bolt Diameter	Bnd	30.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.9088 cm ²
Saddle Load QO (Weight)	QO	2397.7 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	8.5 Kg
Maximum Transverse Force	Ft	102.0 Kg
Maximum Longitudinal Force	Fl	32.6 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 32.6 / (3.909 \cdot 2)$$

$$= 0.4 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 102.0 / (3.909 \cdot 2)$$

$$= 1.3 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 32.56 / (103.2 \cdot 2.0)$$

$$= 0.0155 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 750.0 \cdot 102.01 + 0.0$$

$$= 76.51 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 76.51/2397.74$$

$$= 31.91 \text{ mm.} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 0.0155, 0.0]$$

$$= 0.0155 \text{ cm}^2$$

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: TS IN NOZ. T1

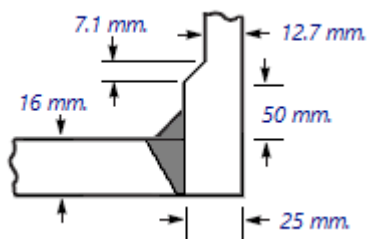
From: 20

Pressure for Reinforcement Calculations	P	25.065	bars
Temperature for Internal Pressure	Temp	120	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	660.00	mm.
Shell Finished (Minimum) Thickness	t	16.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		300.00	mm.
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-90.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.7000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	50.0000	mm.
Height of Beveled Transition	L'	7.1014	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		25.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		16.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: TS IN NOZ. T1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 8.000 in.
Actual Thickness Used in Calculation 0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $25.06 \cdot 333.0 / (138 \cdot 1.0 - 0.6 \cdot 25.06)$
= 6.1153 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $25.06 \cdot 104.6 / (138 \cdot 1.0 - 0.6 \cdot 25.06)$
= 1.9209 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / $\tan(30)$
= $12.3 / 0.5773$
= 21.3042 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

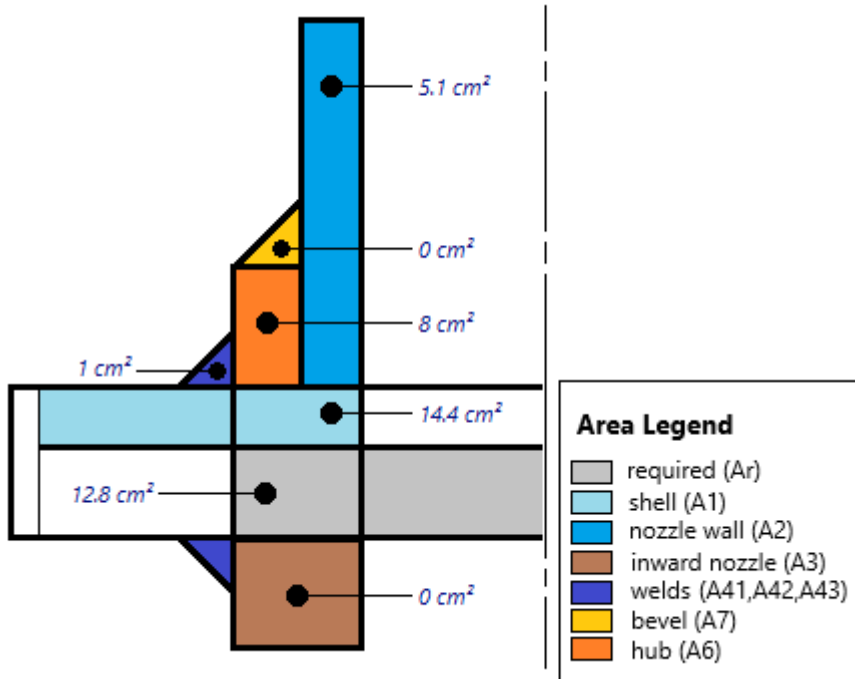
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	418.4000	mm.
Parallel to Vessel Wall, opening length	d	209.2000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm.

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1.0, 1.0)$
= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	12.793	NA	NA
Area in Shell	A1	14.403	NA	NA
Area in Nozzle Wall	A2	5.056	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	7.995	NA	NA
TOTAL AREA AVAILABLE	Atot	28.454	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (209.2 \cdot 6.1153 \cdot 1.0 + 2 \cdot 22.0 \cdot 6.1153 \cdot 1.0 \cdot (1 - 1.0))$
 $= 12.793 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 209.2(1.0 \cdot 13.0 - 1.0 \cdot 6.115) - 2 \cdot 22.0$
 $(1.0 \cdot 13.0 - 1.0 \cdot 6.1153) \cdot (1 - 1.0)$
 $= 14.403 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 32.5)(9.7 - 1.92)1.0$
 $= 5.056 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 1.000 \text{ cm}^2$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(32.5, 200.0, 50.0)) \cdot (25.0 - 12.7) \cdot 1.0$$

$$= 7.995 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 4.9209 \text{ mm.}$
 Wall Thickness per UG16(b), $tr_{16b} = 5.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $tr_{b1} = 9.1153 \text{ mm.}$
 Wall Thickness $tb_1 = \max(tr_{b1}, tr_{16b}) = 9.1153 \text{ mm.}$
 Wall Thickness $tb_2 = \max(tr_{b2}, tr_{16b}) = 5.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb_3 = 11.1026 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[11.103, \max(9.1153, 5.5)]$$

$$= 9.1153 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(4.9209, 9.1153)$$

$$= 9.1153 \text{ mm.}$$

Available Nozzle Neck Thickness = 12.7000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 36.3,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 308.7 N./mm ²	Passed
Occasional	: 12.9,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 15.5,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 12.7$, $t_r = 1.921$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.198$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 16.0$, $t_r = 6.115$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.47$, Temp. Reduction = 37 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 25.06/102.00 = 0.246$$

Weld Size Calculations, Description: TS IN NOZ. T1

Intermediate Calc. for nozzle/shell Welds $T_{min} \quad 13.0000 \text{ mm.}$



Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (12.7933 - 14.4027 + 2 • 22.0 • 1.0 •
(1.0 • 13.0 - 6.1153))138)
= 1997.87 Kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (13.0514 + 1. - 0.0 • 1.0) • 138
= 19772.54 Kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (13.0514 + 0.0 + 1. + (5.72)) • 138
= 27821.47 Kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (13.0514 + 0.0 + 1. + 0.0 + (5.72)) • 138
= 27821.47 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 253.2 • 10.0 • 0.49 • 138
= 27423. Kgf

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 115.6) • (25.0 - 3.0) • 0.7 • 138
= 78699. Kgf

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.1416/2.0) • 253.2 • (16.0 - 3.0) • 0.74 • 138
= 53840. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (27423 + 78699) = 106123 Kgf

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (27423 + 0 + 53840 + 0) = 81263 Kgf

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (27423 + 53840 + 0) = 81263 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 106122 Kgf, must exceed W = 1997 Kgf	or W1 = 19772 Kgf
Path 2-2 = 81262 Kgf, must exceed W = 1997 Kgf	or W2 = 27821 Kgf
Path 3-3 = 81262 Kgf, must exceed W = 1997 Kgf	or W3 = 27821 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 39 bars

The Drop for this Nozzle is : 25.2502 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 241.2502 mm.

Input Echo, WRC297 Item 1, Description: TS IN NOZ. T1 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	660.000	mm.
Shell Thickness	16.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	203.200	mm.
Nozzle Thickness	25.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	25.06	bars
Radial Load	P	-1223.74	Kgf
Circumferential Shear	Vc	917.81	Kgf
Longitudinal Shear	Vl	1223.74	Kgf
Circumferential Moment	Mc	636.10	Kg-m.
Longitudinal Moment	Ml	636.10	Kg-m.
Torsional Moment	Mt	733.96	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 253.2 + 2 \cdot 1.65 \cdot \sqrt{(339.5(16.0 - 3.0))}$
 $= 472.433 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	679.000	mm.
Nozzle Outside Diameter	(d) =	253.200	mm.
Vessel Thickness used	(T) =	13.000	mm.
Nozzle Thickness used	(t) =	22.000	mm.
T / t	=	0.591	
d / t	=	11.509	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.695$$

Nr/P	=	0.064
Mr/P	=	0.113
M0/P	=	0.036
N0/P	=	0.092



MrD/Mc = 0.264
 NrDL/Mc = 0.104
 M0d/Mc = 0.083
 N0DL/Mc = 0.080
 MrD/Ml = 0.101
 NrDL/Ml = 0.053
 M0D/Ml = 0.030
 N0DL/Ml = 0.110

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	15	-15	15	-15
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-16	-16	16	16
Outplane Bending (ML)	-26	26	26	-26
Normal Pressure Stress	62	65	62	65
Outplane Stress Summary	42	67	127	46

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	48	-48	48	-48
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-88	88	88	-88
Inplane Pressure Stress	31	31	31	31
Inplane Stress Summary	-12	68	180	-93

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5
Shear Stress Summary	7	7	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	56	75	181	139

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	15	-15	15	-15
Outplane Membrane (Mc)	-11	-11	11	11
Outplane Bending (Mc)	-72	72	72	-72
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	31	31	31	31



Outplane Stress Summary -30 83 137 -37

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	48	-48	48	-48
Inplane Membrane (Mc)	-15	-15	15	15
Inplane Bending (Mc)	-230	230	230	-230
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	62	65	62	65
Inplane Stress Summary	-129	237	361	-193

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	5	5	5	5
Shear Stress Summary	3	3	7	7

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	129	237	361	194

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-16	-16	16	16
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	12	15	12	15
Hoop Stress Summary	3	6	35	38

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	12	-12	12	-12
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-7	-7	7	7
Axial Bending (ML)	-22	22	22	-22
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	-10	10	49	-20

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
	Outside	Inside	Outside	Inside
Shear due to (Vc)	1	1	-1	-1

Shear due to (V1)	0	0	0	0
Shear due to Torsion	3	3	3	3
Shear Stress Summary	4	4	2	2

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	16	13	49	59

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-11	-11	11	11
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	12	15	12	15
Hoop Stress Summary	7	10	31	34

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	12	-12	12	-12
Axial Membrane (Mc)	-7	-7	7	7
Axial Bending (Mc)	-64	64	64	-64
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	-52	52	91	-62

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (V1)	-1	-1	1	1
Shear due to Torsion	3	3	3	3
Shear Stress Summary	1	1	4	4

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	60	52	91	96

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	62	65	62	65	62	65	62	65
Circ. Pl (SUS)	-10	-10	22	22	-11	-11	19	19
Circ. Q (SUS)	-11	11	41	-41	-182	182	278	-278
Long. Pm (SUS)	31	31	31	31	31	31	31	31



Long. P1 (SUS)	-3	-3	11	11	-5	-5	17	17
Long. Q (SUS)	-40	40	136	-136	-57	57	87	-87
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-2	-2	2	2
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	62.0	65.0	62.0	65.0	62.0	65.0	62.0	65.0
Pm+P1 (SUS)	52.0	55.0	84.0	87.0	51.2	54.1	81.1	84.1
Pm+P1+Q (Total)	54.3	73.1	178.3	140.2	131.1	236.1	359.2	194.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	65.00	138.00	Passed
Pm+P1 (SUS)	87.02	207.00	Passed
Pm+P1+Q (TOTAL)	359.22	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	12	15	12	15	12	15	12	15
Circ. P1 (SUS)	-10	-10	22	22	-5	-5	17	17
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. P1 (SUS)	-7	-7	7	7	-7	-7	7	7
Long. Q (SUS)	-10	10	34	-34	-52	52	76	-76
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	12.0	15.0	12.0	15.0	12.0	15.0	12.0	15.0
Pm+P1 (SUS)	3.6	6.3	34.0	37.0	8.2	11.2	29.1	32.1
Pm+P1+Q (Total)	15.3	11.5	47.3	58.1	60.1	51.1	89.3	95.3

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	15.00	138.00	Passed
Pm+P1 (SUS)	37.04	207.00	Passed
Pm+P1+Q (TOTAL)	95.34	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

Input, Nozzle Desc: TS OUT NOZ. T2

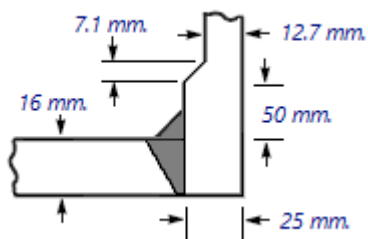
From: 20

Pressure for Reinforcement Calculations	P	25.000	bars
Temperature for Internal Pressure	Temp	120	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	660.00	mm.
Shell Finished (Minimum) Thickness	t	16.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		300.00	mm.
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.7000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	50.0000	mm.
Height of Beveled Transition	L'	7.1014	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		25.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: TS OUT NOZ. T2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 8.000 in.
Actual Thickness Used in Calculation 0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $25.0 \cdot 333.0 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$
= 6.0994 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $25.0 \cdot 104.6 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$
= 1.9159 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / $\tan(30)$
= $12.3 / 0.5773$
= 21.3042 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

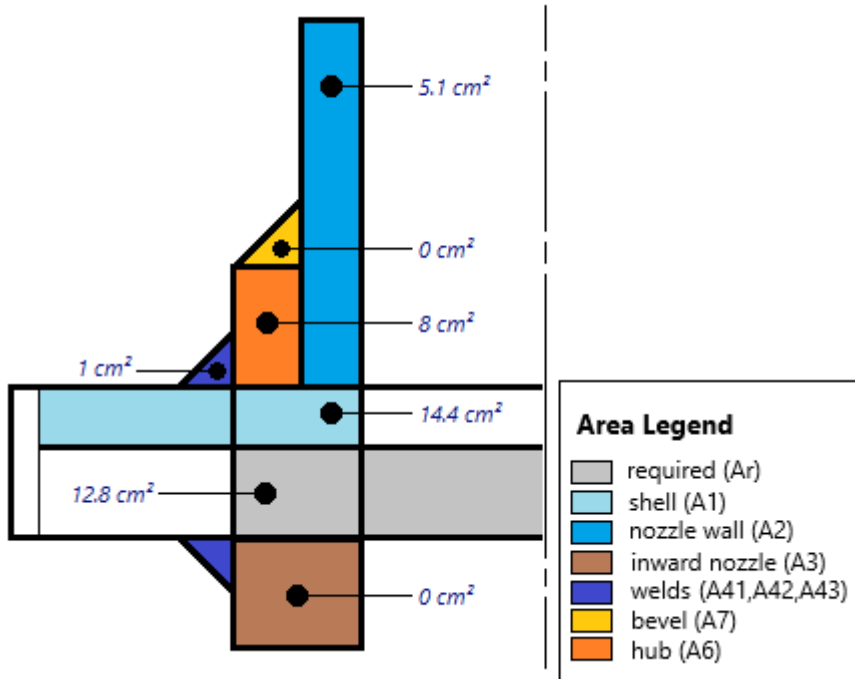
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	418.4000	mm.
Parallel to Vessel Wall, opening length	d	209.2000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm.

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 138.0 / 138.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1.0, 1.0)$
= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	12.760	NA	NA
Area in Shell	A1	14.436	NA	NA
Area in Nozzle Wall	A2	5.060	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	7.995	NA	NA
TOTAL AREA AVAILABLE	Atot	28.491	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (209.2 \cdot 6.0994 \cdot 1.0 + 2 \cdot 22.0 \cdot 6.0994 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 12.760 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 209.2(1.0 \cdot 13.0 - 1.0 \cdot 6.099) - 2 \cdot 22.0$$

$$(1.0 \cdot 13.0 - 1.0 \cdot 6.0994) \cdot (1 - 1.0)$$

$$= 14.436 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 32.5)(9.7 - 1.92)1.0$$

$$= 5.060 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2 \\ &= (2 \cdot \min(32.5, 200.0, 50.0)) \cdot (25.0 - 12.7) \cdot 1.0 \\ &= 7.995 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 4.9159 \text{ mm.}$
Wall Thickness per UG16(b), $tr16b = 5.5000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure $trb1 = 9.0994 \text{ mm.}$
Wall Thickness $tb1 = \max(trb1, tr16b) = 9.0994 \text{ mm.}$
Wall Thickness $tb2 = \max(trb2, tr16b) = 5.5000 \text{ mm.}$
Wall Thickness per table UG-45 $tb3 = 11.1026 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[11.103, \max(9.0994, 5.5)] \\ &= 9.0994 \text{ mm.} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(4.9159, 9.0994) \\ &= 9.0994 \text{ mm.} \end{aligned}$$

Available Nozzle Neck Thickness = 12.7000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	36.3,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	308.7 N./mm ²	Passed
Occasional	:	12.9,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	15.5,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $tg = 12.7$, $tr = 1.916$, $c = 3.0 \text{ mm.}$, $E* = 1.0$
Thickness Ratio = $tr \cdot E*/(tg - c) = 0.198$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $tg = 16.0$, $tr = 6.099$, $c = 3.0 \text{ mm.}$, $E* = 1.0$
Thickness Ratio = $tr \cdot E*/(tg - c) = 0.469$, Temp. Reduction = 37 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 25.00/102.00 = 0.245$$

Weld Size Calculations, Description: TS OUT NOZ. T2

Intermediate Calc. for nozzle/shell Welds $T_{min} \quad 13.0000 \text{ mm.}$



Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (12.7599 - 14.4361 + 2 \cdot 22.0 \cdot 1.0 \cdot (1.0 \cdot 13.0 - 6.0994)) \cdot 138)$
 $= 1913.80 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (13.0547 + 1. - 0.0 \cdot 1.0) \cdot 138$
 $= 19777.12 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (13.0547 + 0.0 + 1. + (5.72)) \cdot 138$
 $= 27826.06 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (13.0547 + 0.0 + 1. + 0.0 + (5.72)) \cdot 138$
 $= 27826.06 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 253.2 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 27423. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 115.6) \cdot (25.0 - 3.0) \cdot 0.7 \cdot 138$
 $= 78699. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 253.2 \cdot (12.0 - 0.0) \cdot 0.74 \cdot 138$
 $= 49698. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (27423 + 78699) = 106123 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (27423 + 0 + 49698 + 0) = 77121 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (27423 + 49698 + 0) = 77121 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 106122 Kgf, must exceed W = 1913 Kgf	or W1 = 19777 Kgf
Path 2-2 = 77121 Kgf, must exceed W = 1913 Kgf	or W2 = 27826 Kgf
Path 3-3 = 77121 Kgf, must exceed W = 1913 Kgf	or W3 = 27826 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 39 bars



The Drop for this Nozzle is : 25.2502 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 241.2502 mm.

Input Echo, WRC297 Item 1, Description: TS OUT NOZ. T2:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	660.000	mm.
Shell Thickness	16.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	203.200	mm.
Nozzle Thickness	25.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUS tained loads.*

Design Internal Pressure	Dp	25.00	bars
Radial Load	P	-1223.74	Kgf
Circumferential Shear	Vc	917.81	Kgf
Longitudinal Shear	Vl	1223.74	Kgf
Circumferential Moment	Mc	636.10	Kg-m.
Longitudinal Moment	Ml	636.10	Kg-m.
Torsional Moment	Mt	733.96	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 253.2 + 2 \cdot 1.65 \cdot \sqrt{(339.5(16.0 - 3.0))}$
 $= 472.433 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	679.000	mm.
Nozzle Outside Diameter	(d) =	253.200	mm.
Vessel Thickness used	(T) =	13.000	mm.
Nozzle Thickness used	(t) =	22.000	mm.
T / t	=	0.591	
d / t	=	11.509	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.695$$

Nr/P	=	0.064
Mr/P	=	0.113
M0/P	=	0.036
N0/P	=	0.092



MrD/Mc = 0.264
 NrDL/Mc = 0.104
 MØd/Mc = 0.083
 NØDL/Mc = 0.080
 MrD/Ml = 0.101
 NrDL/Ml = 0.053
 MØD/Ml = 0.030
 NØDL/Ml = 0.110

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	15	-15	15	-15
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-16	-16	16	16
Outplane Bending (ML)	-26	26	26	-26
Normal Pressure Stress	62	65	62	65
Outplane Stress Summary	42	66	126	46

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	48	-48	48	-48
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-88	88	88	-88
Inplane Pressure Stress	31	31	31	31
Inplane Stress Summary	-12	68	180	-93

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5
Shear Stress Summary	7	7	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress	56	74	180	139

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	15	-15	15	-15
Outplane Membrane (Mc)	-11	-11	11	11
Outplane Bending (Mc)	-72	72	72	-72
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	31	31	31	31

Outplane Stress Summary -30 83 137 -37

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	48	-48	48	-48
Inplane Membrane (Mc)	-15	-15	15	15
Inplane Bending (Mc)	-230	230	230	-230
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	62	65	62	65
Inplane Stress Summary	-130	236	361	-194

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	5	5	5	5
Shear Stress Summary	3	3	7	7

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	130	237	361	194

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-16	-16	16	16
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	12	15	12	15
Hoop Stress Summary	3	6	35	38

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	12	-12	12	-12
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-7	-7	7	7
Axial Bending (ML)	-22	22	22	-22
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	-10	10	49	-20

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
	Outside	Inside	Outside	Inside
Shear due to (Vc)	1	1	-1	-1



Shear due to (V1)	0	0	0	0
Shear due to Torsion	3	3	3	3
Shear Stress Summary	4	4	2	2

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	16	13	49	59

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-11	-11	11	11
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	12	15	12	15
Hoop Stress Summary	7	10	31	34

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	12	-12	12	-12
Axial Membrane (Mc)	-7	-7	7	7
Axial Bending (Mc)	-64	64	64	-64
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	-52	52	91	-62

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (V1)	-1	-1	1	1
Shear due to Torsion	3	3	3	3
Shear Stress Summary	1	1	4	4

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	60	52	91	96

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	62	65	62	65	62	65	62	65
Circ. Pl (SUS)	-10	-10	22	22	-11	-11	19	19
Circ. Q (SUS)	-11	11	41	-41	-182	182	278	-278
Long. Pm (SUS)	31	31	31	31	31	31	31	31



Long. P1 (SUS)	-3	-3	11	11	-5	-5	17	17
Long. Q (SUS)	-40	40	136	-136	-57	57	87	-87
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-2	-2	2	2
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	62.0	65.0	62.0	65.0	62.0	65.0	62.0	65.0
Pm+P1 (SUS)	52.0	55.0	84.0	87.0	51.2	54.1	81.1	84.1
Pm+P1+Q (Total)	54.3	73.1	178.3	140.2	131.1	236.1	359.2	194.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	65.00	138.00	Passed
Pm+P1 (SUS)	87.02	207.00	Passed
Pm+P1+Q (TOTAL)	359.22	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	12	15	12	15	12	15	12	15
Circ. P1 (SUS)	-10	-10	22	22	-5	-5	17	17
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. P1 (SUS)	-7	-7	7	7	-7	-7	7	7
Long. Q (SUS)	-10	10	34	-34	-52	52	76	-76
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	12.0	15.0	12.0	15.0	12.0	15.0	12.0	15.0
Pm+P1 (SUS)	3.6	6.3	34.0	37.0	8.2	11.2	29.1	32.1
Pm+P1+Q (Total)	15.3	11.5	47.3	58.1	60.1	51.1	89.3	95.3

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	15.00	138.00	Passed
Pm+P1 (SUS)	37.04	207.00	Passed
Pm+P1+Q (TOTAL)	95.34	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



Input, Nozzle Desc: SS IN NOZ. S1

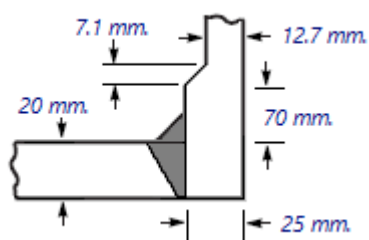
From: 50

Pressure for Reinforcement Calculations	P	32.565	bars
Temperature for Internal Pressure	Temp	145	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	660.00	mm.
Shell Finished (Minimum) Thickness	t	20.0000	mm.
Shell Internal Corrosion Allowance	c	2.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1238.17	mm.
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-90.00	deg
Diameter		16.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.7000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm.
Height of Beveled Transition	L'	7.1014	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		25.0000	mm.
Corrosion Allowance	can	2.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		16.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: SS IN NOZ. S1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 16.000 in.
 Actual Thickness Used in Calculation 0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 32.56 \cdot 332.0 / (138 \cdot 1.0 - 0.6 \cdot 32.56)$
 $= 7.9475$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 32.56 \cdot 205.2 / (138 \cdot 1.0 - 0.6 \cdot 32.56)$
 $= 4.9122$ mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
 $= 2.5 \cdot \text{Corroded Hub Thickness}$
 $= 2.5 \cdot 23.0$, less than the hub height, use (e-2)
 $= 57.5000$ mm.

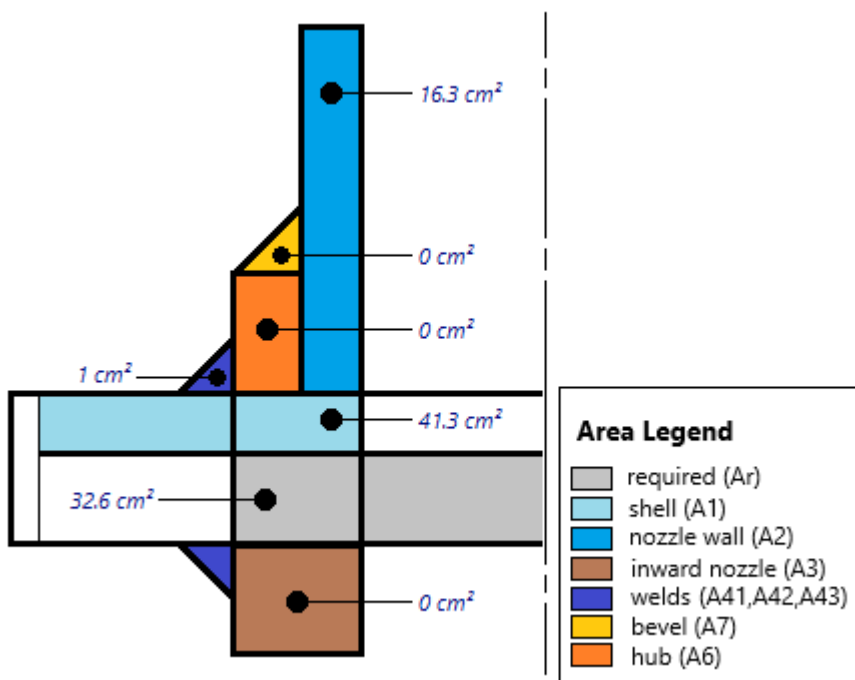
UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	820.8000 mm.
Parallel to Vessel Wall, opening length	d	410.4000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	45.0000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$





Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	32.617	NA	NA
Area in Shell	A1	41.255	NA	NA
Area in Nozzle Wall	A2	16.279	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	58.534	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (410.4 \cdot 7.9475 \cdot 1.0 + 2 \cdot 23.0 \cdot 7.9475 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 32.617 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 410.4(1.0 \cdot 18.0 - 1.0 \cdot 7.948) - 2 \cdot 23.0$$

$$(1.0 \cdot 18.0 - 1.0 \cdot 7.9475) \cdot (1 - 1.0)$$

$$= 41.255 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 45.0)(23.0 - 4.91)1.0$$

$$= 16.279 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

There are no hub area calculations because Figure UG-40 (e-2) was used.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 6.9122 \text{ mm}$.
 Wall Thickness per UG16(b), $tr16b = 3.5000 \text{ mm}$.
 Wall Thickness, shell/head, internal pressure $trb1 = 9.9475 \text{ mm}$.
 Wall Thickness $tb1 = \max(trb1, tr16b) = 9.9475 \text{ mm}$.
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.5000 \text{ mm}$.
 Wall Thickness per table UG-45 $tb3 = 10.3312 \text{ mm}$.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[10.331, \max(9.9475, 3.5)]$$

$$= 9.9475 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(6.9122, 9.9475)$$

$$= 9.9475 \text{ mm}$$

Available Nozzle Neck Thickness = 12.7000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	53.0,	Allowable	:	138.0 N./mm²	Passed
Expansion	:	0.0,	Allowable	:	292.0 N./mm²	Passed



Occasional : 30.4, Allowable : 183.5 N./mm² Passed
 Shear : 14.5, Allowable : 96.6 N./mm² Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 12.7, tr = 4.912, c = 2.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.459, Temp. Reduction = 38 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -1 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -39 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 20.0, tr = 7.948, c = 2.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.442, Temp. Reduction = 42 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -39 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 32.56/51.00 = 0.639

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
 = 2.5 • Corroded Hub Thickness
 = 2.5 • 23.0, less than the hub height, use (e-2)
 = 57.5000 mm.

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 615.6000 mm.
 Parallel to Vessel Wall, opening length d 307.8000 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 45.0000 mm.

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure]	AR	21.746	21.746
Area Available in Shell	A1	20.628	20.628
Area Available in Nozzle Wall	A2	16.279	16.279
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	1.000	1.000
Area Available in Pad	A5	0.000	0.000
Area Available in Hub	A6	0.000	0.000
TOTAL AREA AVAILABLE	Atot	37.907	37.907

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: SS IN NOZ. S1

Intermediate Calc. for nozzle/shell Welds Tmin 18.0000 mm.



Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (32.6167 - 41.2553 + 2 \cdot 23.0 \cdot 1.0 \cdot (1.0 \cdot 18.0 - 7.9475)) \cdot 138) \\
 &= \max(0, -5649.02) \text{ Kgf}
 \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2 + A4 - (W_i - \text{Can} / .707)^2 \cdot f_{r2}) \cdot S_v \\
 &= (16.2791 + 1. - 0.0 \cdot 1.0) \cdot 138 \\
 &= 24314.35 \text{ Kgf}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot (\text{Hub Thickness}) \cdot t \cdot f_{r1})) \cdot S_v \\
 &= (16.2791 + 0.0 + 1. + (8.28)) \cdot 138 \\
 &= 35965.61 \text{ Kgf}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot (\text{Hub Thickness}) \cdot t \cdot f_{r1})) \cdot S_v \\
 &= (16.2791 + 0.0 + 1. + 0.0) \cdot 138 \\
 &= 35965.61 \text{ Kgf}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.1416/2.0) \cdot 456.4 \cdot 10.0 \cdot 0.49 \cdot 138 \\
 &= 49432. \text{ Kgf}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.1416 \cdot 216.7) \cdot (25.0 - 2.0) \cdot 0.7 \cdot 138 \\
 &= 154233. \text{ Kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng} \\
 &= (3.1416/2.0) \cdot 456.4 \cdot (16.0 - 0.0) \cdot 0.74 \cdot 138 \\
 &= 119443. \text{ Kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (49432 + 154233) = 203665 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\
 &= (49432 + 0 + 119443 + 0) = 168874 \text{ Kgf}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\
 &= (49432 + 119443 + 0) = 168874 \text{ Kgf}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 203664 Kgf, must exceed W = 0 Kgf	or W1 = 24314 Kgf
Path 2-2 = 168874 Kgf, must exceed W = 0 Kgf	or W2 = 35965 Kgf
Path 3-3 = 168874 Kgf, must exceed W = 0 Kgf	or W3 = 35965 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:



Converged Maximum Allowable Pressure in the Operating case: 44 bars

The Drop for this Nozzle is : 91.6205 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 311.6205 mm.

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: SS IN NOZ. S1 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	2.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	660.000	mm.
Shell Thickness	20.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	2.0000	mm.
Nozzle Diameter	406.400	mm.
Nozzle Thickness	25.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	32.56	bars
Radial Load	P	-2447.48	Kgf
Circumferential Shear	Vc	1835.61	Kgf
Longitudinal Shear	Vl	2447.48	Kgf
Circumferential Moment	Mc	2544.40	Kg-m.
Longitudinal Moment	Ml	2544.40	Kg-m.
Torsional Moment	Mt	2935.84	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Dv (.652) is > 0.5
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 456.4 + 2 \cdot 1.65 \cdot \sqrt{(341.0(20.0 - 2.0))}$$

$$= 714.940 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	682.000 mm.
Nozzle Outside Diameter	(d) =	456.400 mm.
Vessel Thickness used	(T) =	18.000 mm.
Nozzle Thickness used	(t) =	23.000 mm.
T / t	=	0.783
d / t	=	19.843

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.119$$

Nr/P	=	0.048
Mr/P	=	0.077
M0/P	=	0.028
N0/P	=	0.101
MrD/Mc	=	0.243
NrDL/Mc	=	0.084
M0d/Mc	=	0.082
N0DL/Mc	=	0.102
MrD/M1	=	0.089
NrDL/M1	=	0.041
M0D/M1	=	0.030
N0DL/M1	=	0.125

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-21	-21	21	21
Outplane Bending (ML)	-30	30	30	-30
Normal Pressure Stress	58	61	58	61
Outplane Stress Summary	27	66	129	47

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	34	-34	34	-34
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-90	90	90	-90
Inplane Pressure Stress	29	29	29	29
Inplane Stress Summary	-30	82	163	-84

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	6	6	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	58	84	164	132

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	7	7	7	7



Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	-17	-17	17	17
Outplane Bending (Mc)	-82	82	82	-82
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	29	29	29	29

Outplane Stress Summary	-50	89	148	-41
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	34	-34	34	-34
Inplane Membrane (Mc)	-14	-14	14	14
Inplane Bending (Mc)	-245	245	245	-245
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	58	61	58	61

Inplane Stress Summary	-163	262	355	-200
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	3	3	6	6
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	163	262	356	200
------------------------	-----	-----	-----	-----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-21	-21	21	21
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	33	36	33	36

Hoop Stress Summary	19	23	61	65
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	14	-14	14	-14
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-7	-7	7	7
Axial Bending (ML)	-42	42	42	-42
Axial Pressure Stress	16	16	16	16



Axial Stress Summary -18 37 81 -31

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	3	3	3	3
Shear Stress Summary	4	4	2	2

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	39	39	82	97

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-17	-17	17	17
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	33	36	33	36
Hoop Stress Summary	23	27	57	61

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	14	-14	14	-14
Axial Membrane (Mc)	-7	-7	7	7
Axial Bending (Mc)	-124	124	124	-124
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	16	16	16	16
Axial Stress Summary	-100	119	163	-113

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	3	3	3	3
Shear Stress Summary	2	2	5	5

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	123	119	163	175

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int.	Stress Values at
------------------------	------------------

Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	58	61	58	61	58	61	58	61
Circ. Pl (SUS)	-14	-14	28	28	-11	-11	17	17
Circ. Q (SUS)	-18	18	42	-42	-211	211	279	-279
Long. Pm (SUS)	29	29	29	29	29	29	29	29
Long. Pl (SUS)	-3	-3	9	9	-10	-10	24	24
Long. Q (SUS)	-56	56	124	-124	-70	70	94	-94
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	58.0	61.0	58.0	61.0	58.0	61.0	58.0	61.0
Pm+Pl (SUS)	44.1	47.0	86.0	89.0	47.0	50.0	75.0	78.0
Pm+Pl+Q (Total)	56.9	83.4	162.3	133.1	164.1	261.1	354.1	201.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	61.00	138.00	Passed
Pm+Pl (SUS)	89.02	207.00	Passed
Pm+Pl+Q (TOTAL)	354.12	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	33	36	33	36	33	36	33	36
Circ. Pl (SUS)	-14	-14	28	28	-10	-10	24	24
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	16	16	16	16	16	16	16	16
Long. Pl (SUS)	-7	-7	7	7	-7	-7	7	7
Long. Q (SUS)	-28	28	56	-56	-110	110	138	-138
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	33.0	36.0	33.0	36.0	33.0	36.0	33.0	36.0
Pm+Pl (SUS)	19.1	22.1	61.0	64.0	23.1	26.1	57.0	60.0
Pm+Pl+Q (Total)	38.8	38.0	79.2	97.1	124.1	119.0	161.2	175.2

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	36.00	138.00	Passed
Pm+Pl (SUS)	64.02	207.00	Passed
Pm+Pl+Q (TOTAL)	175.18	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: SS OUT NOZ. S2

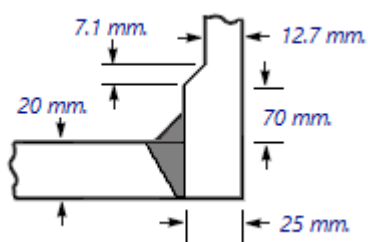
From: 50

Pressure for Reinforcement Calculations	P	32.500	bars
Temperature for Internal Pressure	Temp	145	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	660.00	mm.
Shell Finished (Minimum) Thickness	t	20.0000	mm.
Shell Internal Corrosion Allowance	c	2.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		5138.17	mm.
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		16.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.7000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm.
Height of Beveled Transition	L'	7.1014	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		25.0000	mm.
Corrosion Allowance	can	2.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		16.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: SS OUT NOZ. S2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 16.000 in.
 Actual Thickness Used in Calculation 0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 32.5 \cdot 332.0 / (138 \cdot 1.0 - 0.6 \cdot 32.5)$
 $= 7.9315$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 32.5 \cdot 205.2 / (138 \cdot 1.0 - 0.6 \cdot 32.5)$
 $= 4.9023$ mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
 $= 2.5 \cdot \text{Corroded Hub Thickness}$
 $= 2.5 \cdot 23.0$, less than the hub height, use (e-2)
 $= 57.5000$ mm.

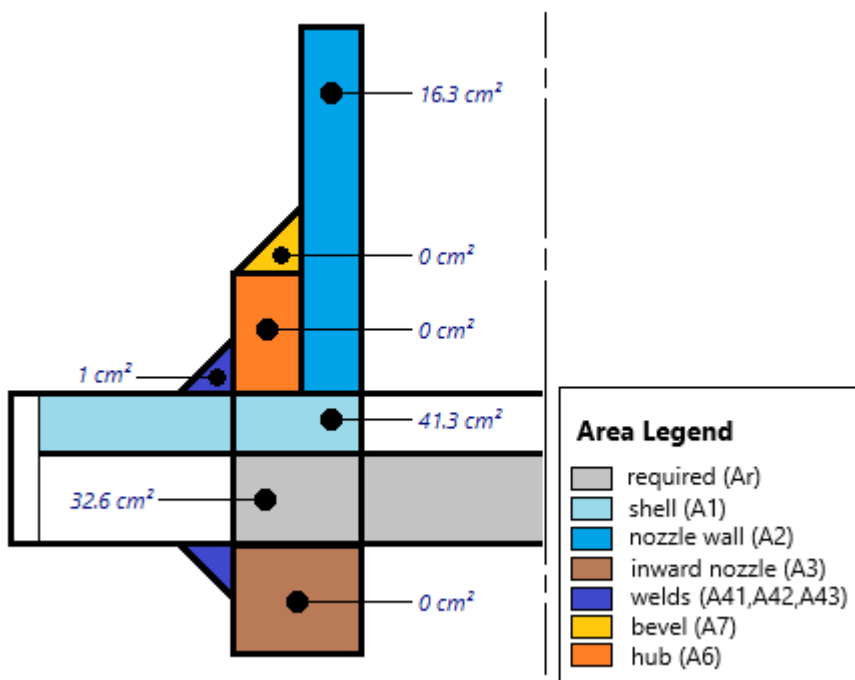
UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	820.8000 mm.
Parallel to Vessel Wall, opening length	d	410.4000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	45.0000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n/S_v)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n/S_v)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$





Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	32.551	NA	NA
Area in Shell	A1	41.321	NA	NA
Area in Nozzle Wall	A2	16.288	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	58.609	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (410.4 \cdot 7.9315 \cdot 1.0 + 2 \cdot 23.0 \cdot 7.9315 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 32.551 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 410.4(1.0 \cdot 18.0 - 1.0 \cdot 7.932) - 2 \cdot 23.0$$

$$(1.0 \cdot 18.0 - 1.0 \cdot 7.9315) \cdot (1 - 1.0)$$

$$= 41.321 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 45.0)(23.0 - 4.9)1.0$$

$$= 16.288 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

There are no hub area calculations because Figure UG-40 (e-2) was used.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 6.9023 \text{ mm}$.
 Wall Thickness per UG16(b), $tr16b = 3.5000 \text{ mm}$.
 Wall Thickness, shell/head, internal pressure $trb1 = 9.9315 \text{ mm}$.
 Wall Thickness $tb1 = \max(trb1, tr16b) = 9.9315 \text{ mm}$.
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.5000 \text{ mm}$.
 Wall Thickness per table UG-45 $tb3 = 10.3312 \text{ mm}$.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[10.331, \max(9.9315, 3.5)]$$

$$= 9.9315 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(6.9023, 9.9315)$$

$$= 9.9315 \text{ mm}$$

Available Nozzle Neck Thickness = 12.7000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	52.9,	Allowable	:	138.0 N./mm²	Passed
Expansion	:	0.0,	Allowable	:	292.1 N./mm²	Passed



Occasional : 30.4, Allowable : 183.5 N./mm² Passed
 Shear : 14.5, Allowable : 96.6 N./mm² Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 12.7, tr = 4.902, c = 2.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.458, Temp. Reduction = 39 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -1 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -40 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 20.0, tr = 7.932, c = 2.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.441, Temp. Reduction = 42 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -40 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 32.50/51.00 = 0.637

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
 = 2.5 • Corroded Hub Thickness
 = 2.5 • 23.0, less than the hub height, use (e-2)
 = 57.5000 mm.

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 615.6000 mm.
 Parallel to Vessel Wall, opening length d 307.8000 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 45.0000 mm.

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure]	AR	21.702	21.702
Area Available in Shell	A1	20.661	20.661
Area Available in Nozzle Wall	A2	16.288	16.288
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	1.000	1.000
Area Available in Pad	A5	0.000	0.000
Area Available in Hub	A6	0.000	0.000
TOTAL AREA AVAILABLE	Atot	37.948	37.948

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: SS OUT NOZ. S2

Intermediate Calc. for nozzle/shell Welds Tmin 18.0000 mm.



Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (32.551 - 41.321 + 2 \cdot 23.0 \cdot 1.0 \cdot (1.0 \cdot 18.0 - 7.9315)) \cdot 138) \\
 &= \max(0, -5823.62) \text{ Kgf}
 \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2 + A4 - (W_i - \text{Can} / .707)^2 \cdot f_{r2}) \cdot S_v \\
 &= (16.288 + 1. - 0.0 \cdot 1.0) \cdot 138 \\
 &= 24326.89 \text{ Kgf}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot (\text{Hub Thickness}) \cdot t \cdot f_{r1})) \cdot S_v \\
 &= (16.288 + 0.0 + 1. + (8.28)) \cdot 138 \\
 &= 35978.15 \text{ Kgf}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot (\text{Hub Thickness}) \cdot t \cdot f_{r1})) \cdot S_v \\
 &= (16.288 + 0.0 + 1. + 0.0) \cdot 138 \\
 &= 35978.15 \text{ Kgf}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.1416/2.0) \cdot 456.4 \cdot 10.0 \cdot 0.49 \cdot 138 \\
 &= 49432. \text{ Kgf}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.1416 \cdot 216.7) \cdot (25.0 - 2.0) \cdot 0.7 \cdot 138 \\
 &= 154233. \text{ Kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng} \\
 &= (3.1416/2.0) \cdot 456.4 \cdot (16.0 - 0.0) \cdot 0.74 \cdot 138 \\
 &= 119443. \text{ Kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (49432 + 154233) = 203665 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\
 &= (49432 + 0 + 119443 + 0) = 168874 \text{ Kgf}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\
 &= (49432 + 119443 + 0) = 168874 \text{ Kgf}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 203664 Kgf, must exceed W = 0 Kgf	or W1 = 24326 Kgf
Path 2-2 = 168874 Kgf, must exceed W = 0 Kgf	or W2 = 35978 Kgf
Path 3-3 = 168874 Kgf, must exceed W = 0 Kgf	or W3 = 35978 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:



Converged Maximum Allowable Pressure in the Operating case: 44 bars

The Drop for this Nozzle is : 91.6205 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 261.6205 mm.

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: SS OUT NOZ. S2:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	2.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	660.000	mm.
Shell Thickness	20.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	2.0000	mm.
Nozzle Diameter	406.400	mm.
Nozzle Thickness	25.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	32.50	bars
Radial Load	P	-2447.48	Kgf
Circumferential Shear	Vc	1835.61	Kgf
Longitudinal Shear	Vl	2447.48	Kgf
Circumferential Moment	Mc	2544.40	Kg-m.
Longitudinal Moment	Ml	2544.40	Kg-m.
Torsional Moment	Mt	2935.84	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Dv (.652) is > 0.5
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 456.4 + 2 \cdot 1.65 \cdot \sqrt{(341.0(20.0 - 2.0))}$$

$$= 714.940 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	682.000	mm.
Nozzle Outside Diameter	(d) =	456.400	mm.
Vessel Thickness used	(T) =	18.000	mm.
Nozzle Thickness used	(t) =	23.000	mm.
T / t	=	0.783	
d / t	=	19.843	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.119$$

Nr/P	=	0.048
Mr/P	=	0.077
M0/P	=	0.028
N0/P	=	0.101
MrD/Mc	=	0.243
NrDL/Mc	=	0.084
M0d/Mc	=	0.082
N0DL/Mc	=	0.102
MrD/M1	=	0.089
NrDL/M1	=	0.041
M0D/M1	=	0.030
N0DL/M1	=	0.125

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-21	-21	21	21
Outplane Bending (ML)	-30	30	30	-30
Normal Pressure Stress	58	61	58	61
Outplane Stress Summary	26	65	129	47

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	34	-34	34	-34
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-90	90	90	-90
Inplane Pressure Stress	29	29	29	29
Inplane Stress Summary	-30	82	163	-84

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	6	6	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	58	84	164	132

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	7	7	7	7



Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	-17	-17	17	17
Outplane Bending (Mc)	-82	82	82	-82
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	29	29	29	29

Outplane Stress Summary	-50	89	148	-41
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	34	-34	34	-34
Inplane Membrane (Mc)	-14	-14	14	14
Inplane Bending (Mc)	-245	245	245	-245
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	58	61	58	61

Inplane Stress Summary	-163	262	355	-200
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	3	3	6	6
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	163	262	356	200
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-21	-21	21	21
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	32	36	32	36

Hoop Stress Summary	19	23	61	65
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	14	-14	14	-14
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-7	-7	7	7
Axial Bending (ML)	-42	42	42	-42
Axial Pressure Stress	16	16	16	16



Axial Stress Summary -18 37 81 -31

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	3	3	3	3
Shear Stress Summary	4	4	2	2

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	39	39	82	97

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-17	-17	17	17
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	32	36	32	36
Hoop Stress Summary	23	27	57	61

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	14	-14	14	-14
Axial Membrane (Mc)	-7	-7	7	7
Axial Bending (Mc)	-124	124	124	-124
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	16	16	16	16
Axial Stress Summary	-100	119	163	-113

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	3	3	3	3
Shear Stress Summary	2	2	5	5

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	123	119	163	175

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int.	Stress Values at
------------------------	------------------

Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	58	61	58	61	58	61	58	61
Circ. Pl (SUS)	-14	-14	28	28	-11	-11	17	17
Circ. Q (SUS)	-18	18	42	-42	-211	211	279	-279
Long. Pm (SUS)	29	29	29	29	29	29	29	29
Long. Pl (SUS)	-3	-3	9	9	-10	-10	24	24
Long. Q (SUS)	-56	56	124	-124	-70	70	94	-94
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	58.0	61.0	58.0	61.0	58.0	61.0	58.0	61.0
Pm+Pl (SUS)	44.1	47.0	86.0	89.0	47.0	50.0	75.0	78.0
Pm+Pl+Q (Total)	56.9	83.4	162.3	133.1	164.1	261.1	354.1	201.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	61.00	138.00	Passed
Pm+Pl (SUS)	89.02	207.00	Passed
Pm+Pl+Q (TOTAL)	354.12	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	32	36	32	36	32	36	32	36
Circ. Pl (SUS)	-14	-14	28	28	-10	-10	24	24
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	16	16	16	16	16	16	16	16
Long. Pl (SUS)	-7	-7	7	7	-7	-7	7	7
Long. Q (SUS)	-28	28	56	-56	-110	110	138	-138
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	32.0	36.0	32.0	36.0	32.0	36.0	32.0	36.0
Pm+Pl (SUS)	18.1	22.1	60.0	64.0	22.1	26.1	56.0	60.0
Pm+Pl+Q (Total)	37.9	38.0	79.2	97.1	123.1	119.0	161.2	175.2

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	36.00	138.00	Passed
Pm+Pl (SUS)	64.02	207.00	Passed
Pm+Pl+Q (TOTAL)	175.18	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

Input Echo, Tubesheet Number 1, Description: Tubesheet

Shell Data:

Main Shell Description: MAIN SHELL

Shell Maximum Design Pressure	Psd,max	32.50 bars
Shell Maximum Operating Pressure	Psox,max	32.50 bars
Shell Minimum Operating Pressure	Psox,min	0.00 bars
Shell Thickness	ts	20.0000 mm.
Shell Internal Corrosion Allowance	cas	2.0000 mm.
Shell External Corrosion Allowance	casext	0.0000 mm.
Inside Diameter of Shell	Ds	660.000 mm.
Shell Temperature for Internal Pressure	Ts	145.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: CH. SHELL

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	25.00 bars
Channel Maximum Operating Pressure	Ptox,max	25.00 bars
Channel Minimum Operating Pressure	Ptox,min	0.00 bars
Channel Thickness	tc	16.0000 mm.
Channel Internal Corrosion Allowance	cac	3.0000 mm.
Channel External Corrosion Allowance	cacext	0.0000 mm.
Inside Diameter of Channel	Dc	660.000 mm.
Channel Design Temperature	TEMPC	120.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	442
Tube Wall Thickness	et	2.1100 mm.
Tube Outside Diameter	D	19.0500 mm.
Total Straight Tube Length	Lt	4600.00 mm.
Straight Tube Length (bet. inner tubsht faces) L		4515.00 mm.
Design Temperature of the Tubes		145.00 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		92.40 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	158.60 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.4000 mm.
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm.
Groove Weld Leg	ag	2.1100 mm.
Tube-Tubesheet Joint Weld Type	Full	Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	313.500 mm.
Largest Center-to-Center Tube Distance	Ul	57.1500 mm.
Length of Expanded Portion of Tube	ltx	0.0000 mm.
Tube-side pass partition groove depth	hg	6.0000 mm.

Tubesheet Data:



Tubesheet TYPE: U-tube, Gasketed both Sides, Conf. d

Tubesheet Design Metal Temperature T 145.00 °C
 Tubesheet Material SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	85.0000 mm.
Tubesheet Corr. Allowance (Shell side)	Cats	2.0000 mm.
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm.
Tubesheet Outside Diameter	A	835.000 mm.

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		No
Step 1 Diameter on the Tubeside	dt1	724.00 mm.
Step 1 Depth on the Tubeside	ht1	6.00 mm.
Step 2 Diameter on the Tubeside	dt2	657.00 mm.
Step 2 Depth on the Tubeside	ht2	6.00 mm.
Is the Tubesheet Flat on Shellside?		No
Step 1 Diameter on the Shellside	ds1	724.00 mm.
Step 1 Depth on the Shellside	hs1	6.00 mm.
Step 2 Diameter on the Shellside	ds2	657.00 mm.
Step 2 Depth on the Shellside	hs2	6.00 mm.
Calculated Tubesheet Diameter per Div 2, 4.18.4		724.00 mm.

Note: Tubesheet diameter is now: 724.000 mm. Div 2, 4.18.4.

Dimension G for the Channel Side	Gc	705.151 mm.
Area of the Untubed Lanes	AL	369.2 cm ²

Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Both
Flange Outside Diameter	A	835.000 mm.
Flange Inside Diameter	B	660.000 mm.
Flange Face Outside Diameter	Fod	727.000 mm.
Flange Face Inside Diameter	Fid	660.000 mm.
Gasket Outside Diameter	Go	718.000 mm.
Gasket Inside Diameter	Gi	692.000 mm.
Small end Hub thk.	g0	20.0000 mm.
Large end Hub thk.	g1	25.0000 mm.
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	68.95 N./mm ²
Flange Facing Sketch	Code Sketch 1a	
Gasket Thickness	tg	4.5000 mm.
Full face Gasket Flange Option	Program Selects	
Length of Partition Gasket	lp	646.000 mm.
Width of Partition Gasket	wp	13.0000 mm.
Partition Gasket Factor,	mPart	3.7500
Partition Gasket Design Seating Stress	yPart	62.06 N./mm ²
Partition Gasket Facing Sketch	Code Sketch 1a	
Partition Gasket Column for Gasket Seating	Code Column II	

Bolting Information:

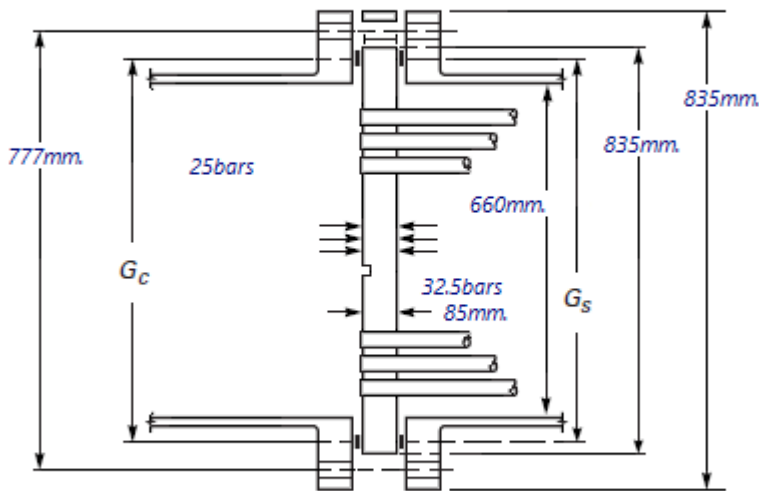
Diameter of Bolt Circle	C	777.000 mm.
Nominal Bolt Diameter	dB	24.0000 mm.
Thread Series	TEMA Metric Thread	
Number of Bolts	n	32
Bolt Material	SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm.



Alternate Flange Operating Bolt Load, Wm1 168381.67 Kgf
 Alternate Flange Seating Bolt Load, Wm2 126326.55 Kgf
 Alternate Flange Design Bolt Load, W 175523.97 Kgf

Tubesheet Integral with None
 Tubesheet Extended as Flange Yes
 Thickness of Extended Portion of Tubesheet Tf 61.0000 mm.
 Is Bolt Load Transferred to the Tubesheet Yes

Tubesheet Gasketed w/Shell and Channel



Configuration d:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	145.0 °C	0.19536E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	120.0 °C	0.19691E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Tubsht. Design Temp.	145.0 °C	0.19536E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	145.0 °C	0.19536E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (25.0 \cdot (19.05/2 - 0)) / (92.4 \cdot 1.0 + 0.4 \cdot 25.0)$
 $= 0.2550 + 0.0000 = 0.2550 \text{ mm.}$

Tube Required Thickness under External Pressure (Shellside pressure) :



External Pressure Chart CS-1 at 145.00 °C
Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	4828.50	9.03	50.0000	0.0134948	95.84

EMAWP = (2.167/(D/T)-0.0833)*B = 150.192 bars

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.5985	19.05	4828.50	31.83	50.0000	0.0010857	77.60

EMAWP = (4*B)/(3*(D/T)) = (4 *77.596)/(3 *31.8299) = 32.5026 bars

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.5985 mm.
Allowable Internal Pressure at Corroded thickness	224.57 bars
Required Internal Design Pressure	25.00 bars
Allowable External Pressure at Corroded thickness	150.19 bars
Required External Design Pressure	32.50 bars
Required Thickness due to Shell Side pressure	0.5985 mm.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (G_{oc} - G_{ic}) / 2$	13.000 mm.
Basic Gasket Width,	$b_0 = N / 2$	6.500 mm.
Effective Gasket Width,	$b = \sqrt{b_0} * 2.5$	6.422 mm.
Gasket Reaction Diameter,	$G = G_o - 2.0 * b$	705.156 mm.

Results for ASME U-tube Tubesheet Calculations for Configuration d:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$= 2 * r_o + d_t$
 $= 2 * 313.5 + 19.05$
 $= 646.050 \text{ mm.}$

Basic ligament efficiency for shear [μ]:

$= (p - d_t) / p$
 $= (25.4 - 19.05) / 25.4$
 $= 0.250$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ρ_s]:

$= G_s / D_o$ (Configurations d, e, f)
 $= 705.1556 / 646.05$
 $= 1.0915$

Ratio [ρ_c]:

$= G_c / D_o$ (Configurations d)
 $= 705.1509 / 646.05$
 $= 1.0915$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$= l_{tx} / h$
 $= 0 / 85.0$
 $= 0.000$ (must be $0 \leq \rho \leq 1$)



Effective tube hole Diameter [d*]:
 $= \max(dt - 2tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2tt)$
 $= \max(19.05 - 2 \cdot 2.11 (0.19536E+09/.19536E+09) \cdot$
 $(92/137) \cdot (0); 19.05 - 2 \cdot 2.11)$
 $= 19.0500 \text{ mm.}$

Effective tube pitch [p*]:
 $= p / \sqrt{ (1 - 4 \cdot \min(A; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$
 $= 25.4 / \sqrt{ (1 - 4 \cdot \min(369.19; 4 \cdot 646.05 \cdot 25.4) / (\pi \cdot 646.05^2)) }$
 $= 26.9637 \text{ mm.}$

Effective ligament efficiency for bending [μ^*]:
 $= (p^* - d^*) / p^*$
 $= (26.9637 - 19.05) / 26.9637$
 $= 0.2935$

$h/p = 3.346457$; $\mu^* = 0.293495$; $E^* = 53262328 \text{ KPa.}$
 $E^*/E = 0.272638$; $v^* = 0.358419$

Note: As h/p (3.346) is > 2, data values for $h/p = 2$ were used.

Skip Step 4 for Configuration d :

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:
 $= A / Do$
 $= 724.0/646.05$
 $= 1.1207$

Coefficient [F]:
 $= (1 - v^*) / E^* \cdot (E \cdot \ln(K))$
 $= (1 - 0.36) / 53262328 \cdot (0.19536E+09 \cdot \ln(1.12))$
 $= 0.2681$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:
 $= Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt]$
 $= 646.05^2/16 \cdot [(1.091 - 1) \cdot (1.091^2 + 1) \cdot 32.5 -$
 $(1.091 - 1) \cdot (1.091^2 + 1) \cdot 25.0]$
 $= 39234.8633 \text{ bars} \cdot \text{mm.}^2$

Moment acting on unperforated tubesheet rim [M*]:
 $= Mts + W^* \cdot (Gc - Gs) / (2 \cdot \pi \cdot Do)$
 $= 39234.9 + 168381.7 \cdot (705.151 - 705.151) / (2 \cdot \pi \cdot 646.05)$
 $= 39234.8633 \text{ bars} \cdot \text{mm.}^2$

Note: W^* is the maximum of the bolt loads between the shell and channel sides.

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:
 $= ((M^*) - Do^2/32 \cdot F(Ps - Pt)) / (1 + F)$
 $= ((39234.86) - 646.05^2/32 \cdot 0.268 (32.5 - 25.0)) / (1 + 0.27)$
 $= 10260.9307 \text{ bars} \cdot \text{mm.}^2$

Maximum bending moment acting on center of tubesheet [Mo]:
 $= Mp + Do^2/64 \cdot (3 + v^*) \cdot (Ps - Pt)$
 $= 10260.93 + 646.05^2/64 \cdot (3 + 0.358) \cdot (32.5 - 25.0)$
 $= 174527.2188 \text{ bars} \cdot \text{mm.}^2$

Maximum bending moment acting on tubesheet [M]:
 $= \max(| Mp |; | Mo |)$
 $= \max(| 10260.931 |; | 174527.219 |)$
 $= 174527.2188 \text{ bars} \cdot \text{mm.}^2$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:



$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 174527.219 / (0.2935 (85.0 - 6.0)^2) \\ &= 57.1722 \text{ N./mm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned} &= 2 \cdot S \\ &= 2 \cdot 138.0 \\ &= 275.99 \text{ N./mm}^2 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 174472.203 / (0.2935 (41.9663 - 6.0)^2) \\ &= 275.7475 \text{ N./mm}^2 \end{aligned}$$

Required tubesheet thickness for Bending Stress [H_{reqB}]:

$$\begin{aligned} &= h + C_{ats} + C_{atc} \\ &= 41.9663 + 0 + 0 \\ &= 41.9663 \text{ mm.} \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [H_{req}]:

$$\begin{aligned} &= \max(H_{reqB}; H_{reqS}) \\ &= \max(41.9663 ; 4.3893) \\ &= 41.9663 \text{ mm.} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$| P_s - P_t |$]:

$$\begin{aligned} &= | P_s - P_t | \\ &= | 32.5 - 25.0 | \\ &= 7.500 \text{ bars} \end{aligned}$$

Shear Stress check [τ_{limit}]:

$$\begin{aligned} &= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y) \\ &= 4 \cdot 0.25 \cdot 85.0 / 646.05 \cdot \min(0.8 \cdot 137.997 ; 0.533 \cdot 219.798) \\ &= 14.525 \text{ N./mm}^2 \end{aligned}$$

$| P_s - P_t |$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$\begin{aligned} &= 1 / (4 \cdot \mu) \cdot (D_o / \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot | P_s - P_t | + C_{ats} + C_{atc} \\ &= 1 / (4 \cdot 0.25) \cdot (646.05 / 110.398) \cdot | 32.5 - 25.0 | + 0 \\ &= 4.3893 \text{ mm.} \end{aligned}$$

Note: Analysis Completed for Tubesheet Configuration d.

Tube Metal Area [A_t]:

$$\begin{aligned} &= \pi (d_o - t) t \\ &= \pi (19.05 - 2.11) 2.11 \\ &= 1.123 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per UW-20.3:

Tube Strength [F_t]:

$$\begin{aligned} &= \pi \cdot t (d_o - t) S \\ &= \pi \cdot 2.11 (19.05 - 2.11) 92.398 \\ &= 1057.989 \text{ Kg} \end{aligned}$$

Groove Weld Strength [F_g]:

$$\begin{aligned} &= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t) \\ &= 0.85 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 92.4 \\ &= 1057.9891 \text{ Kg} \end{aligned}$$

Weld Strength Factor [f_w]:

$$\begin{aligned} &= S / S_w \text{ where } S_w = \min(S \text{ and } S_t) \\ &= 92.398 / 92.398 \\ &= 1.000 \end{aligned}$$

Design Strength Ratio [f_d]:

$$= 1 \text{ for full strength welds}$$



Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 92.4$$

$$= 702.9343 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd)} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 19.05)^2 + 1.07 \cdot 2.11 \cdot (19.05 - 2.11) \cdot 1.0 \cdot 1.0)} - 0.75 \cdot 19.05 \right)$$

$$= 2.562 \text{ mm.}$$

Minimum Required Fillet Weld Leg [afr]: 1.2810 mm.

Minimum Required Groove Weld Leg [agr]: 1.2810 mm.

Tube-Tubesheet Jt allowable, 1057.99 is >= tube strength 1057.99 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	57.2 <=	276.0 N./mm ²	Ok
Tubesheet shear stress	5.7 <=	110.4 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm.)	Required	Actual	P/F
Tubesheet Thickness :	41.966	85.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.281	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.281	2.110	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	71.738 ...	191	276	19	110 Ps+Pt D1	Ok
D2uc	80.871 34.747	248	276	25	110 Ps+Pt D2	Ok
D3uc	41.966 ...	57	276	6	110 Ps+Pt D3	Ok
D1c	73.664 ...	201	276	20	110 Ps+Pt-c D1	Ok
D2c	82.849 34.747	261	276	26	110 Ps+Pt-c D2	Ok
D3c	43.948 ...	60	276	6	110 Ps+Pt-c D3	Ok
Max:	82.8494 34.747 mm.	0.945	0.238	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Note:

Because there was no net pressure differential across the tubesheet due to



the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm.)	Required	Actual	P/F
Tubesheet thickness :	82.849	85.000	Ok
Tubesheet thickness flanged extension :	34.747	61.000	Ok
Tube thickness :	0.598	2.110	Ok
Tube-tubesheet fillet weld leg :	1.281	2.110	Ok
Tube-tubesheet groove weld leg :	1.281	2.110	Ok

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	34.397	1.000	34.397	1.000
Tubesheet Shear Stress	136.696	1.000	136.696	1.000
Tube Pressure Stress	224.569	1.000	150.191	1.000
Tubesheet Extension Stress	No Calc	No Calc	33.947	...
Minimum MAWP	34.397		33.947	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	36.207	1.000	36.207	1.000
Tubesheet Shear Stress	145.240	1.000	145.240	1.000
Tube Pressure Stress	224.569	1.000	150.191	1.000
Tubesheet Extension Stress	No Calc	No Calc	33.947	...
Minimum MAPnc	36.207		33.947	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Determine the governing MDMT considering the governing condition:

Governing thickness on the shell side per figure UCS-66.3 (c):
 = tubesheet thickness/4
 = 85.0/4
 = 21.250 mm.

Thickness Ratio = 0.957, Temperature Reduction per Fig. UCS 66.1 = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -8 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

where the MDMT reduction ratio per UCS 66 (b)(1)(b) is:
 = max(pt/Tubeside MAPnc, ps/Shellside MAPnc), must be ≤ 1
 = max(25.0/36.21, 32.5/33.95)
 = 0.957

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